MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title:

CALCULATION
TERTIARY CRUSHING
DYNAMIC ANALYSIS OF CRUSHER SUPPORT STRUCTURE

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PREFACE

VIBRATION PROBLEM - TERTIARY CRUSHER (2 NOS.) & SUPPORTING STRUCTURE

A. The Problem:

- Crusher No. 1 & 2 was started with VVFD, and no vibration was observed during the individual run or simultaneous operation of both crushers. However, after one week or so, vibration was observed for both individual operation of crusher 1 & 2 as well as simultaneous operations of both crushers at 50% and 100% of speed.

- Though vibration isolators have been provided below the Crusher, the vibration was being transmitted to the supporting concrete structure and could be felt by all.

- Vibration measurement was taken at different speed of crusher and maximum observed value in horizontal direction was 12 mm/s (RMS) on concrete deck support. ISO 10816 specifies an allowable limit of 4-5 mm/s, and the project specification limit is 3 mm/s. Manufacturer has not specified any allowable limit of vibration.

B. Root Cause Analysis

- For the applied pattern of dynamic loads, the maximum velocity at concrete support was computed around 4 mm/s. However, the actual measured velocity at concrete support was found to be 11.6 mm/s. This indicates that there was a misinterpretation of the forced vibration nature of the equipment, considered in the analysis, with respect to the actual load arising from the operating equipment.

- There was a deviation from project specific limits with respect to being away from the range of excitation frequencies and vibration amplitude limit.

- There was a misinterpretation of the geotechnical advice regarding foundation design parameters for dynamic foundations.

- A revised analysis has been done with a modified load pattern, as confirmed by the equipment manufacturer. The modified loading pattern is presented in Chapter 3 and the confirmation by the equipment manufacturer, is given in Appendix 2. A detailed soil strata modelling has also been done in this analysis.

- With the above modifications in analysis methodology, the computed velocity is 14 mm/s, which approximately matches the actual measured value of 12 m/s at site. However, this is much more than the allowable, comfortable vibration limit of 3 mm/s.

- This is the reason why extreme vibration is being felt on the structure.

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C. Proposed Stiffening of Structure:

- To reduce the vibration level to an allowable limit, additional stiffening of the structure is required, with additional concrete walls, as shown in Appendix 1.
- The modified structure has been analysed with the corrected load pattern and with detailed soil strata modelling.

D. Conclusion:

- The stiffening has reduced the computed vibration to 2.74 mm/s, which is within the acceptable limit.
- Also, as a result of stiffening of the structure, the natural frequency is now 9.4 Hz which is 2.01 times the operating frequency of the equipment.
- The modified structure is adequate for the loads arising from the equipment.

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1. Introduction

This report is a verification of the Tertiary Crusher system, using finite element analysis [1]. The main purpose of this analysis is to investigate:

- Root cause analysis for the excessive vibration of the existing structure. Harmonic analyses are performed considering empty conditions of the crusher.
- Undamped modal behavior to check with the excitation frequencies of the proposed Tertiary Crusher. Two modal analyses are performed considering empty and loaded conditions of the crusher.
- Steady state dynamic behavior of the proposed system with harmonic analysis to ensure that potential excitation forces will not cause vibration that might jeopardize safe operation of the equipment. Two harmonic analyses are performed considering empty and loaded conditions of the crusher.
- The concrete design on modifications is shown in revised Tertiary Crushing Foundations and Superstructures (MD-216-3130-EG-CV-CAL-1007)

2. Design Basis

2.1. Geometry of the Model

Tertiary Crusher system consists of the RCC footing, columns, beams, ties, slab, and different pedestals.

2.1.1. Existing Concrete Structure

Dimension of the existing concrete structure is given in Appendix 1.

2.1.2. Proposed Concrete Structure

Dimension of the proposed concrete structure is given in Appendix 1.

2.2. Material Properties

The material properties of concrete shown in Table 1a.

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Property	Value	Unit
Material Field Variables	Table	
Density	2400	kg m^-3
Isotropic Secant Coefficient of Thermal Expansion		
Isotropic Elasticity		
Derive from	Young's Mod...	
Young's Modulus	3.1528E+10	Pa
Poisson's Ratio	0.17	
Bulk Modulus	1.5923E+10	Pa
Shear Modulus	1.3474E+10	Pa

Table 1a. Material properties of concrete

2.3. Soil Properties

Borehole 19 is near Crusher System. Shear modulus & Poisson ratios are taken from the soil report to model the soil in different layers. The material properties of soil in Figure 1. Dimension of the soil model is considered as 45m X 45m X 15m depth.

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TEST REPORT - DOWNHOLE SEISMIC TEST RESULTS *(ASTM D7400 - 08)*

FORM # RGF-KHO-GEO-DH-1-00

Project: *MANSOURAH MASSARAH GOLD PROJECT*
Client: *L & T CONSTRUCTION*
Project No.: *19-K-3054* **Test No.** *DH-03*
Borehole No.: *BH-19* **Geotechnical File No.:** -
Source Type: *Sledge Hammer* **Source to BH Distance (m):** 3 m
Receiver Type: *Triaxial Geophones* **Coordinates (m):** *N: 2506510.84, E: 264424.58*

Sr. No.	Depth of Reciever (m)	Distance of Source to Receiver (m)	Average Time		Calculated Velocity		Assumed Density	Poisson Ratio	Shear Modulus	Young Modulus	Bulk Modulus
			P-wave (msec)	S-wave (msec)	P-wave (m/sec)	S-wave (m/sec)	ρ (g/cm³)	υ	G (M Pa)	E (M Pa)	K (M Pa)
1	0.75	3.09	5.10	9.09	606	340	1.7	0.27	197	500	363
2	1.50	3.35	5.38	9.56	935	557	1.7	0.22	527	1292	783
3	2.25	3.75	5.78	10.21	990	609	1.8	0.20	668	1596	873
4	3.00	4.24	6.28	11.01	985	616	1.8	0.18	683	1610	837
5	3.75	4.80	6.87	11.97	949	583	1.8	0.20	612	1464	804
6	4.50	5.41	7.48	12.93	993	631	1.8	0.16	717	1666	820
7	5.25	6.05	8.16	14.04	939	575	1.8	0.20	595	1428	793
8	6.00	6.71	8.81	15.09	1018	630	1.8	0.19	714	1699	912
9	6.75	7.39	9.44	16.15	1077	640	1.8	0.23	737	1809	1104
10	7.50	8.08	9.94	16.97	1382	843	1.9	0.20	1350	3250	1830
11	8.25	8.78	10.26	17.50	2190	1322	2.1	0.21	3671	8908	5176
12	9.00	9.49	10.59	18.04	2146	1312	2.2	0.20	3785	9099	5089
13	9.75	10.20	10.90	18.55	2304	1401	2.2	0.21	4315	10417	5926
14	10.50	10.92	11.22	19.08	2247	1357	2.2	0.21	4050	9825	5709
15	11.25	11.64	11.58	19.66	2008	1246	2.2	0.19	3418	8113	4315
16	12.00	12.37	11.90	20.18	2269	1397	2.2	0.20	4291	10256	5609
17	12.75	13.10	12.23	20.74	2209	1302	2.2	0.23	3727	9198	5763
18	13.50	13.83	12.56	21.26	2216	1406	2.2	0.16	4349	10115	5000
19	14.25	14.56	12.96	21.89	1833	1164	2.2	0.16	2979	6924	3417
20	15.00	15.30	13.42	22.63	1597	993	2.2	0.19	2169	5140	2721
21	15.75	16.03	13.87	23.39	1636	969	2.2	0.23	2064	5077	3135
22	16.50	16.77	14.26	24.02	1891	1170	2.2	0.19	3014	7168	3846

Figure 1a. Soil Properties for different layers

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Properties of Outline Row 11: Rock1

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1700	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	5.0038E+08	Pa
9	Poisson's Ratio	0.27	
10	Bulk Modulus	3.6259E+08	Pa
11	Shear Modulus	1.97E+08	Pa

Properties of Outline Row 22: Rock2

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1700	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.2859E+09	Pa
9	Poisson's Ratio	0.22	
10	Bulk Modulus	7.654E+08	Pa
11	Shear Modulus	5.27E+08	Pa

Properties of Outline Row 26: Rock3

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.6032E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	8.9067E+08	Pa
11	Shear Modulus	6.68E+08	Pa

Properties of Outline Row 27: Rock4

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.6119E+09	Pa
9	Poisson's Ratio	0.18	
10	Bulk Modulus	8.3952E+08	Pa
11	Shear Modulus	6.83E+08	Pa

Figure 1b. Properties of soil layers 1 to 4

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Properties of Outline Row 28: Rock5

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.4688E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	8.16E+08	Pa
11	Shear Modulus	6.12E+08	Pa

Properties of Outline Row 29: Rock6

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.6634E+09	Pa
9	Poisson's Ratio	0.16	
10	Bulk Modulus	8.1541E+08	Pa
11	Shear Modulus	7.17E+08	Pa

Properties of Outline Row 30: Rock7

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.428E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	7.9333E+08	Pa
11	Shear Modulus	5.95E+08	Pa

Properties of Outline Row 31: Rock8

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.6993E+09	Pa
9	Poisson's Ratio	0.19	
10	Bulk Modulus	9.1361E+08	Pa
11	Shear Modulus	7.14E+08	Pa

Figure 1c. Properties of soil layers 5 to 8

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Properties of Outline Row 32: Rock9

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1800	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.813E+09	Pa
9	Poisson's Ratio	0.23	
10	Bulk Modulus	1.1191E+09	Pa
11	Shear Modulus	7.37E+08	Pa

Properties of Outline Row 12: Rock10

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	1900	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	3.24E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	1.8E+09	Pa
11	Shear Modulus	1.35E+09	Pa

Properties of Outline Row 13: Rock11

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2100	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	8.8838E+09	Pa
9	Poisson's Ratio	0.21	
10	Bulk Modulus	5.1056E+09	Pa
11	Shear Modulus	3.671E+09	Pa

Properties of Outline Row 14: Rock12

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	9.084E+09	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	5.0467E+09	Pa
11	Shear Modulus	3.785E+09	Pa

Figure 1d. Properties of soil layers 9 to 12

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Properties of Outline Row 15: Rock13

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.0442E+10	Pa
9	Poisson's Ratio	0.21	
10	Bulk Modulus	6.0013E+09	Pa
11	Shear Modulus	4.315E+09	Pa

Properties of Outline Row 16: Rock14

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	9.801E+09	Pa
9	Poisson's Ratio	0.21	
10	Bulk Modulus	5.6328E+09	Pa
11	Shear Modulus	4.05E+09	Pa

Properties of Outline Row 17: Rock15

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	8.1348E+09	Pa
9	Poisson's Ratio	0.19	
10	Bulk Modulus	4.3736E+09	Pa
11	Shear Modulus	3.418E+09	Pa

Properties of Outline Row 18: Rock16

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.0298E+10	Pa
9	Poisson's Ratio	0.2	
10	Bulk Modulus	5.7213E+09	Pa
11	Shear Modulus	4.291E+09	Pa

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Figure 1e. Properties of soil layers 13 to 16

Properties of Outline Row 19: Rock17

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	9.1684E+09	Pa
9	Poisson's Ratio	0.23	
10	Bulk Modulus	5.6595E+09	Pa
11	Shear Modulus	3.727E+09	Pa

Properties of Outline Row 20: Rock18

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	1.009E+10	Pa
9	Poisson's Ratio	0.16	
10	Bulk Modulus	4.9459E+09	Pa
11	Shear Modulus	4.349E+09	Pa

Properties of Outline Row 21: Rock19

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	6.9113E+09	Pa
9	Poisson's Ratio	0.16	
10	Bulk Modulus	3.3879E+09	Pa
11	Shear Modulus	2.979E+09	Pa

Properties of Outline Row 23: Rock20

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	5.1622E+09	Pa
9	Poisson's Ratio	0.19	
10	Bulk Modulus	2.7754E+09	Pa
11	Shear Modulus	2.169E+09	Pa

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Figure 1f. Properties of soil layers 17 to 20

Properties of Outline Row 24: Rock21

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	5.0774E+09	Pa
9	Poisson's Ratio	0.23	
10	Bulk Modulus	3.1342E+09	Pa
11	Shear Modulus	2.064E+09	Pa

Properties of Outline Row 25: Rock22

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2200	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Shear Modulus and Poisson's...	
8	Young's Modulus	7.1733E+09	Pa
9	Poisson's Ratio	0.19	
10	Bulk Modulus	3.8566E+09	Pa
11	Shear Modulus	3.014E+09	Pa

Figure 1g. Properties of soil layers 21 to 22

2.4. Damping Property

Damping of 5% is considered for dynamic analysis, shown in Table 2.

Details of "Analysis Settings"

Options	
Rotordynamics Controls	
Output Controls	
Damping Controls	
Constant Damping Ratio	5.e-002
Stiffness Coefficient Define By	Direct Input
Stiffness Coefficient	0.
Mass Coefficient	0.
Analysis Data Management	

Table 2. Damping properties for harmonic analysis

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2.5. Lump Mass Modelling of Crusher

Mass model of the crusher, listed in Table 2a & 2b, is considered as equivalent amount mass of material for simplification of the model in finite element analysis. Modelling is done taking information from load data given by equipment manufacturer shown A [1,2], B [1,2], and C [1,2] loading as follows:

Loaded Conditions:

- Mass of A1, A2, B1, & B2: 200kN = around 20,000 kg each
- Mass of C1, & C2: 45kN = around 4,500 kg each

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	Static Load	Dynamic Load, normal operation f=4,67 Hz	
Damper position	Vertical	Vertical	Horizontal
A [1, 2]	200 kN	± 13 kN	15 kN
B [1, 2]	200 kN	± 14 kN	16 kN
C [1, 2]	45 kN	± 4 kN	2 kN
Vertical load, positive downwards			
Horizontal load, (all horizontal directions)			

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Table 2a. Load data from [MD - 216 -3120 - EG- ME -GAD -V500-1001 Rev C01] [2]

Empty Conditions:

- Mass of A1, A2, B1, & B2: 145.75kN = around 14,575 kg each
- Mass of C1, & C2: 45kN = around 4,500 kg each

TOTAL WEIGHT	: 58,300 Kg
MAX LENGTH WITH SUBFRAME	: 6,350 mm
WIDTH WITH SUB FRAME	: 3200 mm
HEIGHT	: 4700 mm

Table 2b. Load data from [MD - 216 -3120 - EG- ME -GAD -V500-1001 Rev C01] [2]

The material properties of lumped structures and other components are considered to get such masses and listed in Figure 2a, 2b, & 2c.

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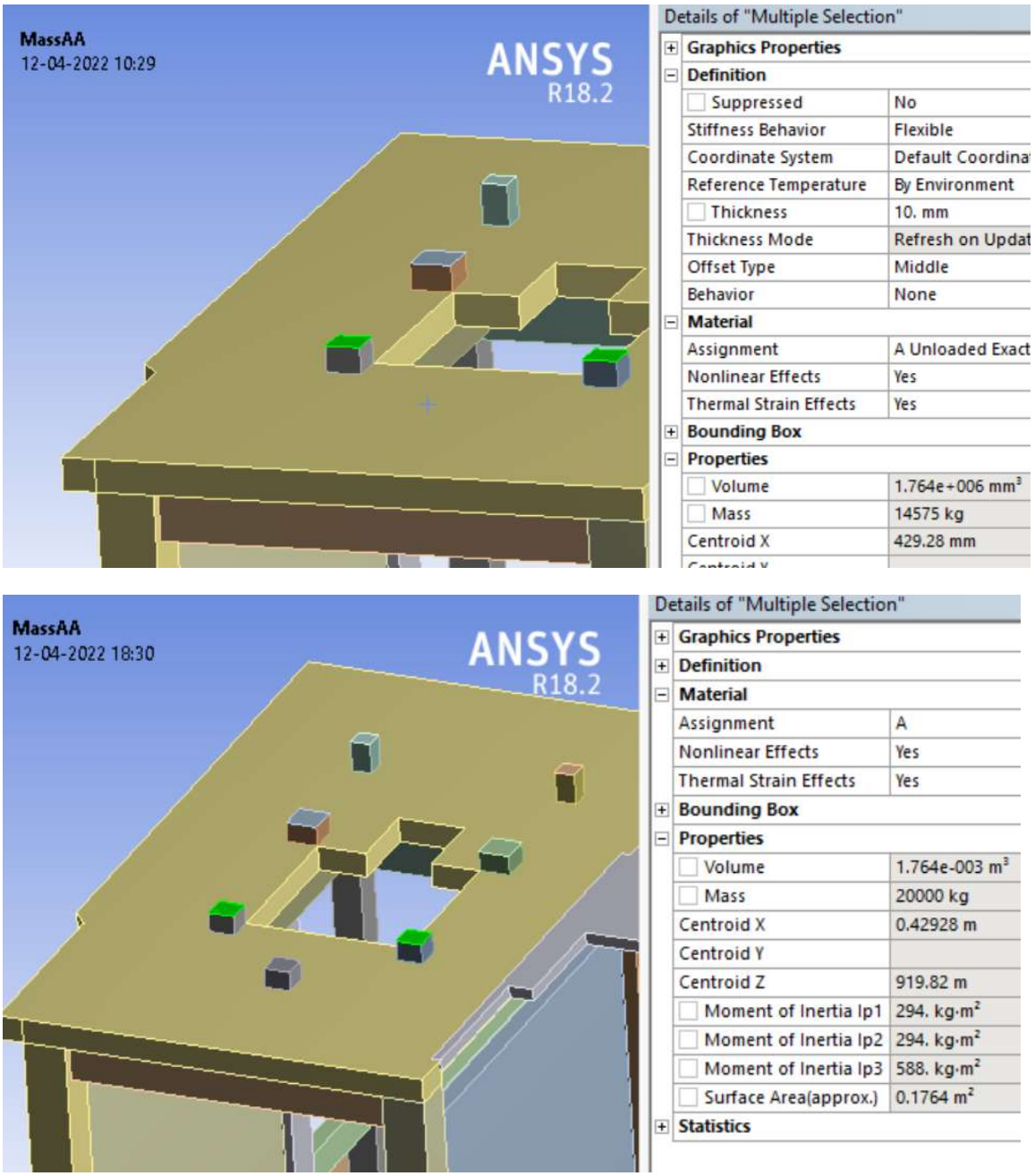


Figure 2a. Lumped mass of the crushers at support: Unloaded & Loaded

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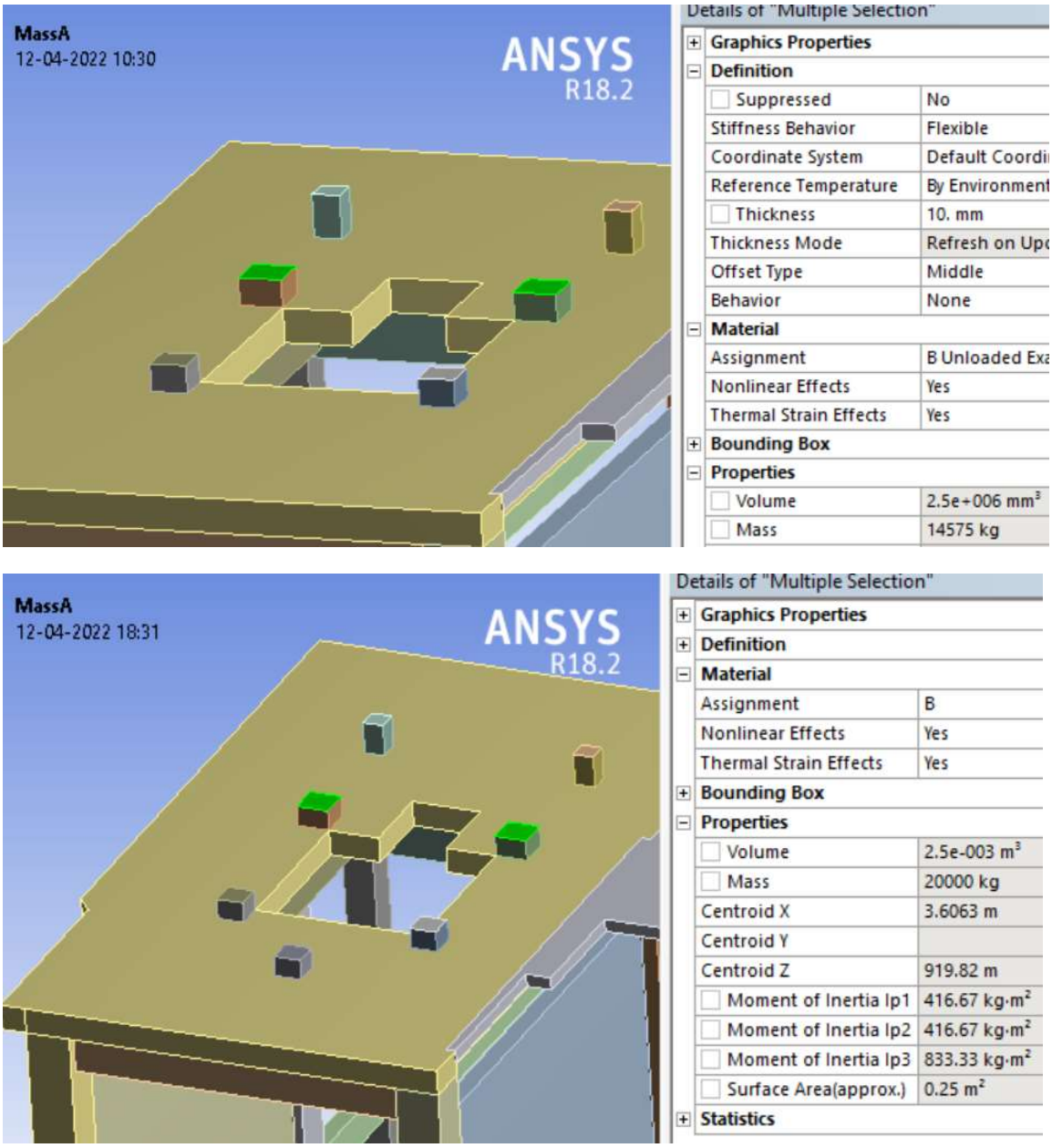


Figure 2b. Lumped mass of the crushers at support: Unloaded & Loaded

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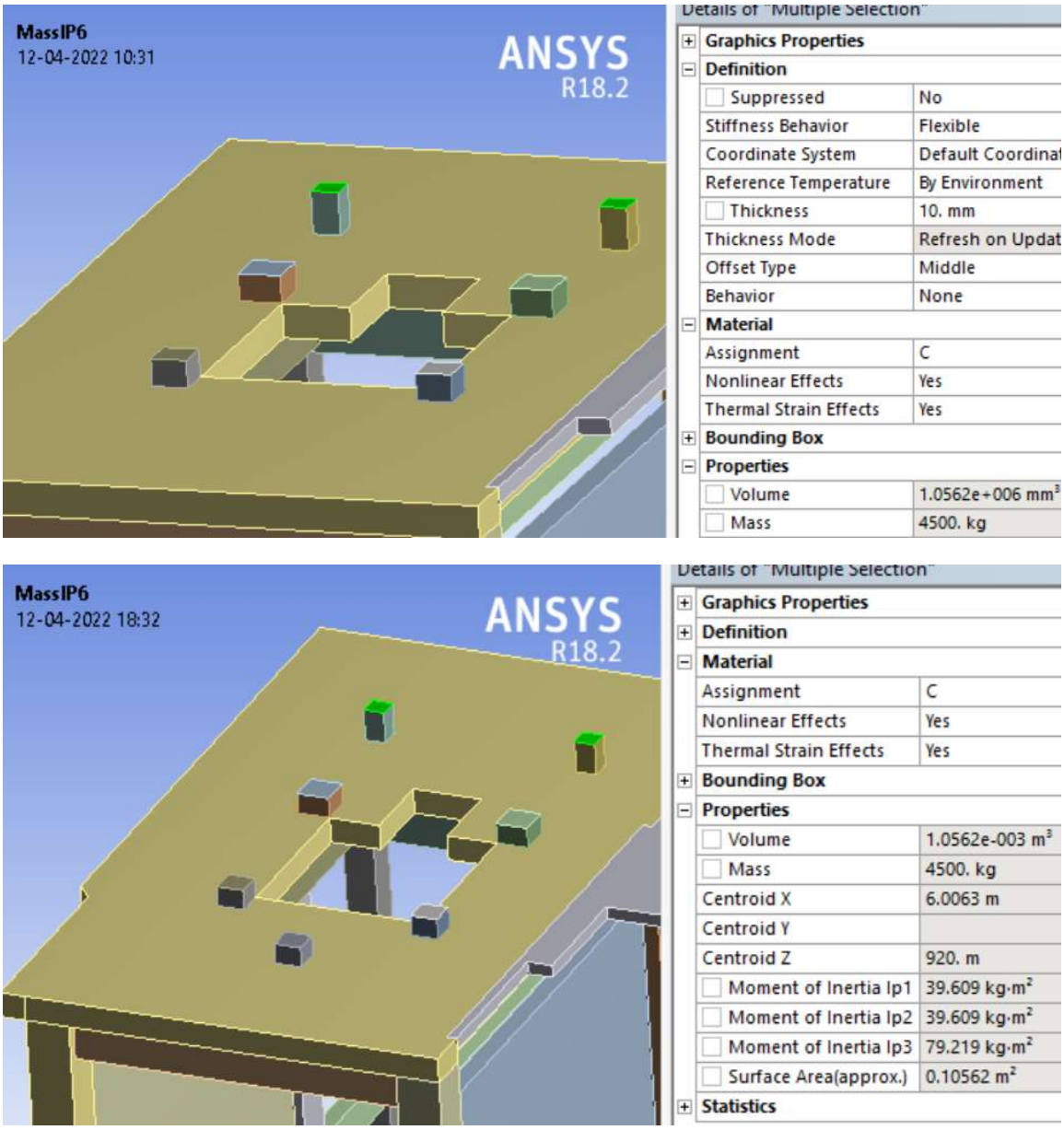


Figure 2c. Lumped mass of the crushers at support: Unloaded & Loaded

3. Modelling of Dynamic Loading

3.1. Dynamic Load, normal operation (f=4.67 Hz)

Components of dynamic forces are taken from drawing provided by equipment manufacturer. Continuous component (Static load) has no effect in dynamic analysis, so ignored in current analysis. Dynamic component is applied in the respective pedestals of the system with different phases and frequency (4.67 Hz).

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3.2. Philosophy of the Operational Dynamic Loadings

All horizontal forces are acting in the same horizontal direction in XY plane as shown in Figure 3a-d. Vertical loads are applied to increase the rocking effect in respective phases.

3.2.1. Time phase 0 Degree

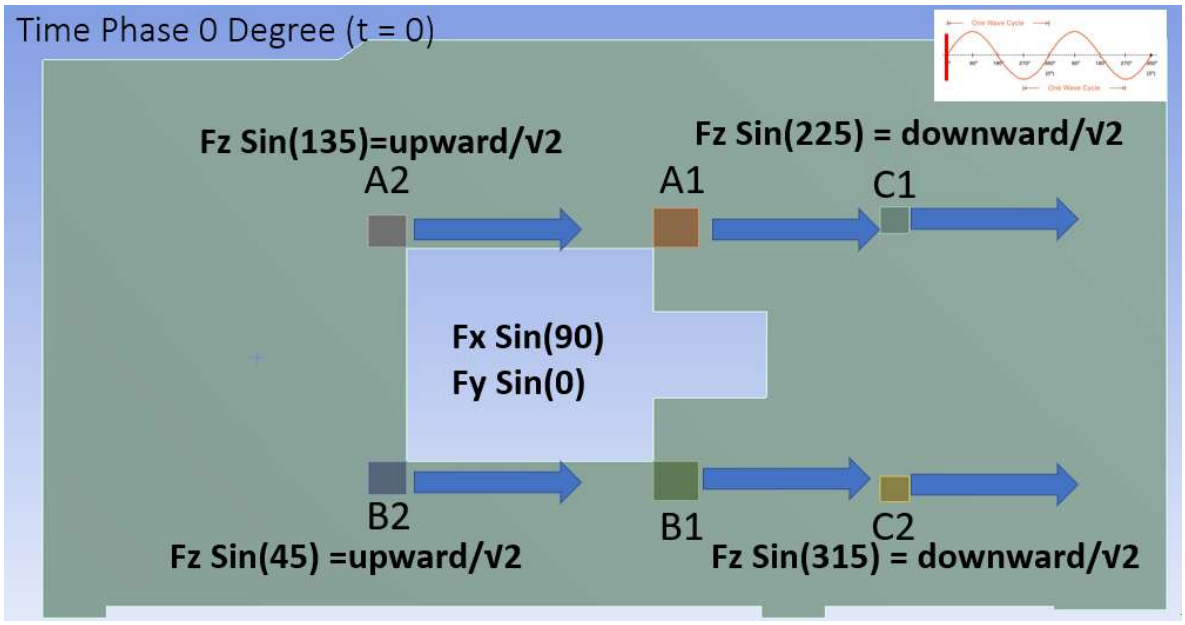


Figure 3a. Excitation Force Direction & Phase angles

3.2.2. Time phase 90 Degree

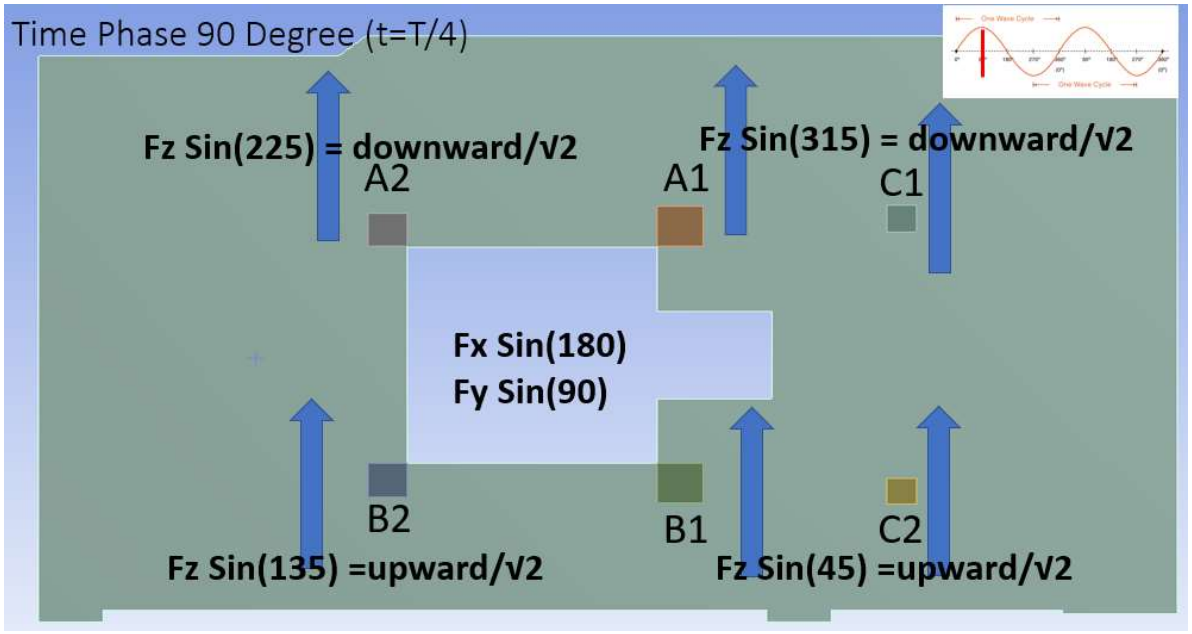


Figure 3b. Excitation Force Direction & Phase angles

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3.2.3. Time phase 180 Degree

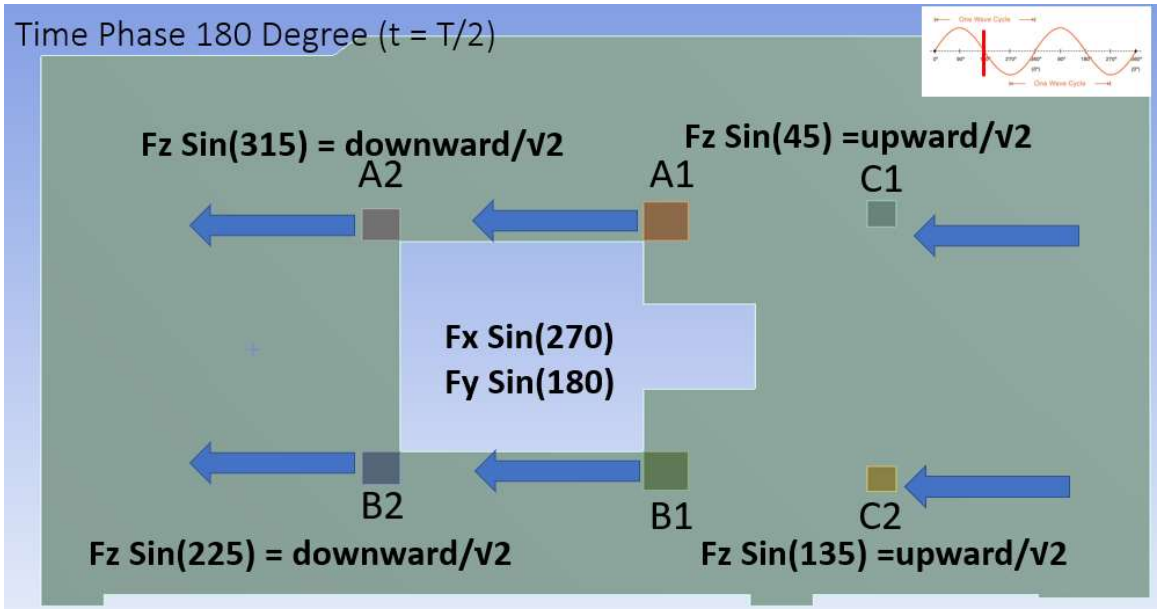


Figure 3c. Excitation Force Direction & Phase angles

3.2.4. Time phase 270 Degree

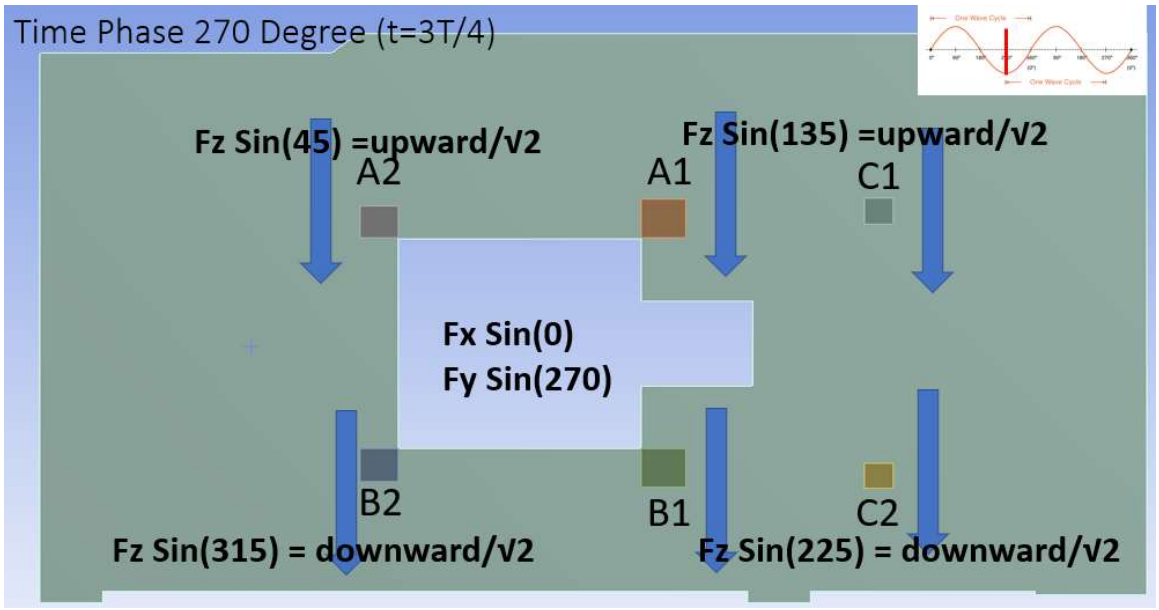


Figure 3d. Excitation Force Direction & Phase angles

3.3. Operational Loading in ANSYS

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As crushers are mirror in mass, stiffness and damping, responses can also be considered as mirror in nature. Responses of the structure for both the crushers running simultaneously will be the summation of the responses of the two crusher systems considering single crusher is running.

3.3.1. Phases of the Loadings in Crusher 1

Following phase angles are given in different pedestals tops:

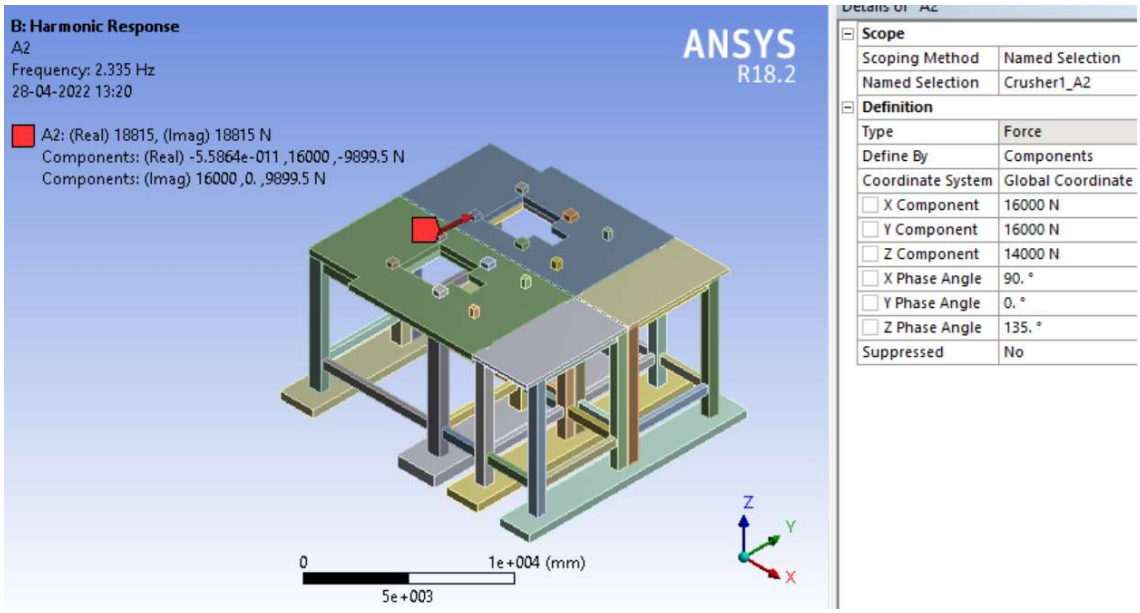


Figure 3a. Excitation Force & Phase angles

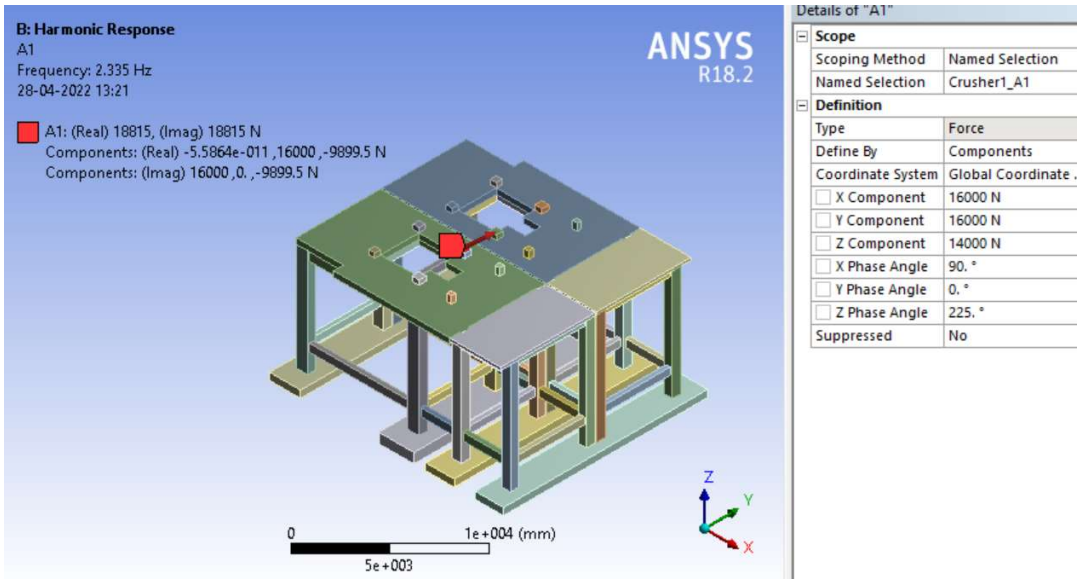


Figure 3b. Excitation Force & Phase angles

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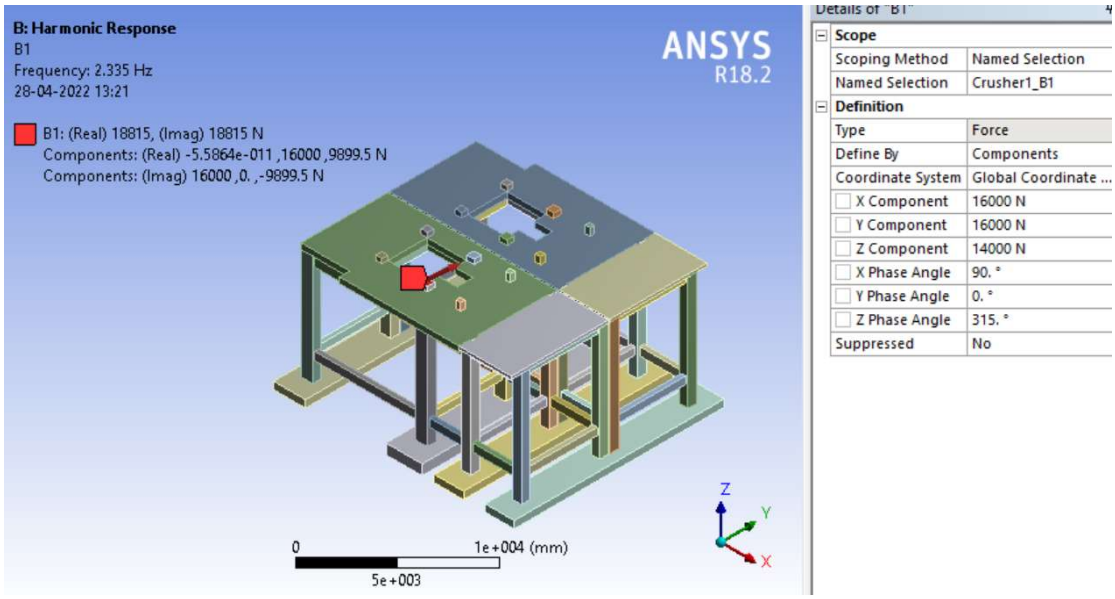


Figure 3c. Excitation Force & Phase angles

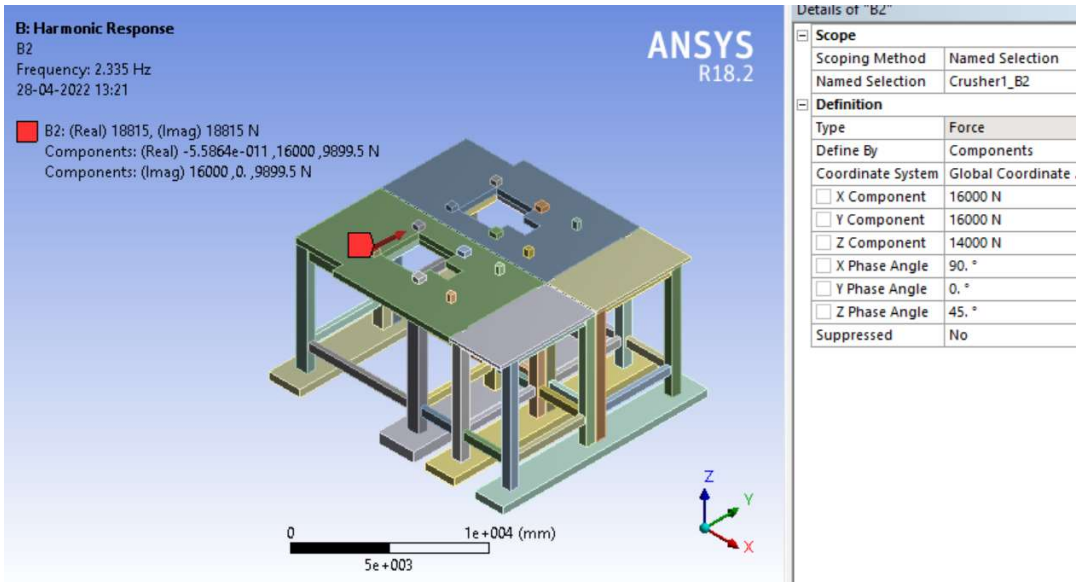


Figure 3d. Excitation Force & Phase angles

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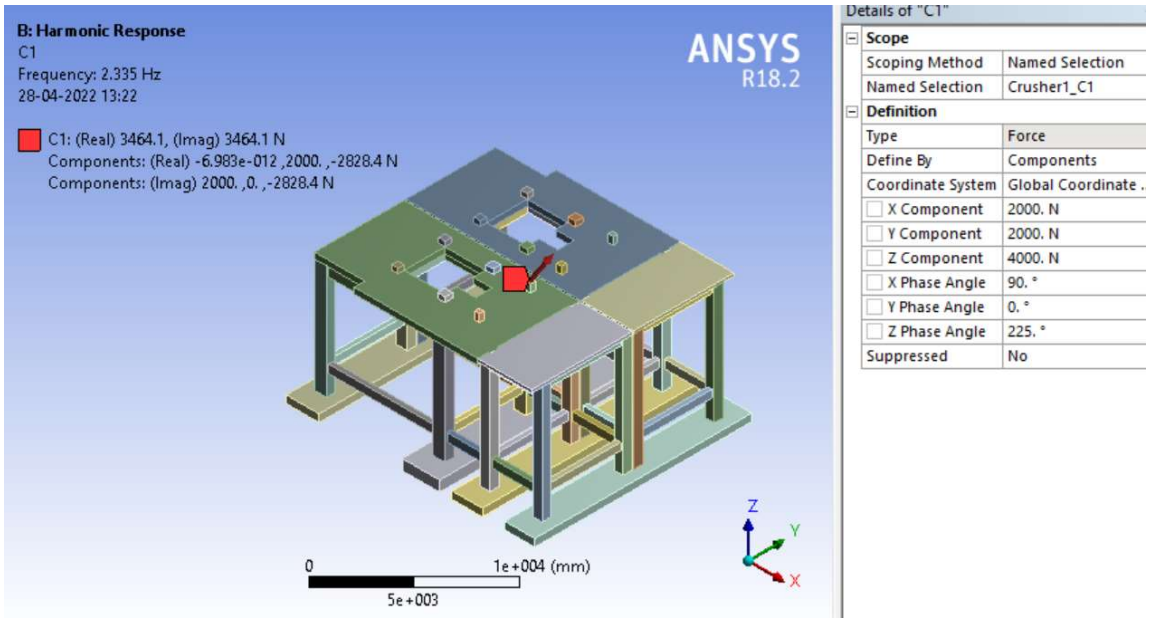


Figure 3e. Excitation Force & Phase angles

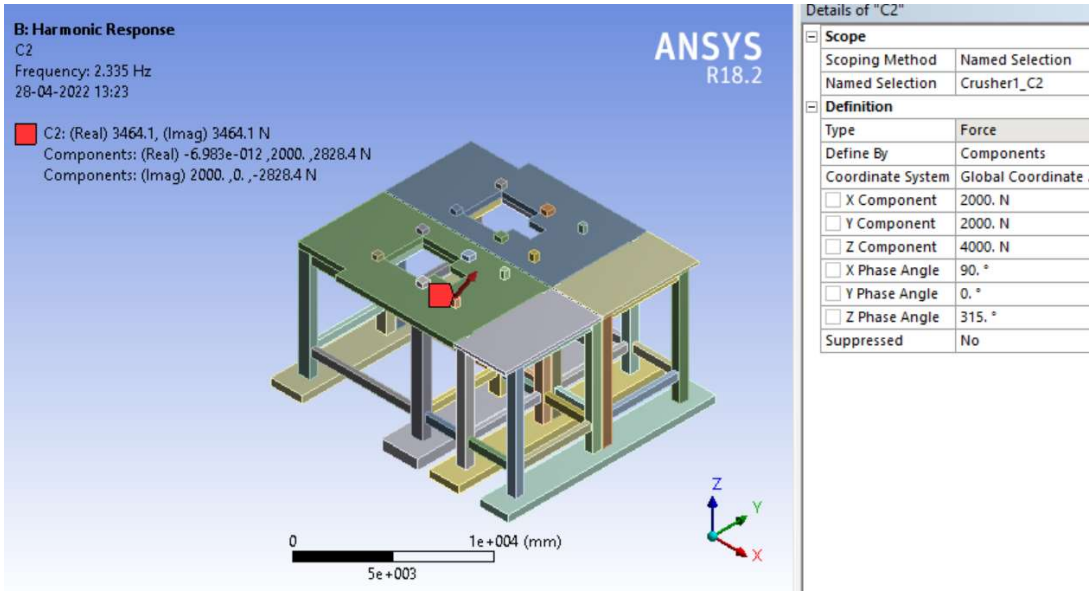


Figure 3f. Excitation Force & Phase angles

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3.4. Dynamic Load, Startup & Shutdown

Dynamic analysis is for the startup & shutdown requires time history analysis, which is computationally extensive. Present report is not addressing startup & shutdown conditions. However, if all the natural frequencies are much above the excitation frequencies, startup & shutdown will be very smooth, which can be identified from the FRF of the structure.

3.5. Boundary Conditions

The model is attached with respective boundary condition at the bottom & sides of the soil, shown in the Figure 4.

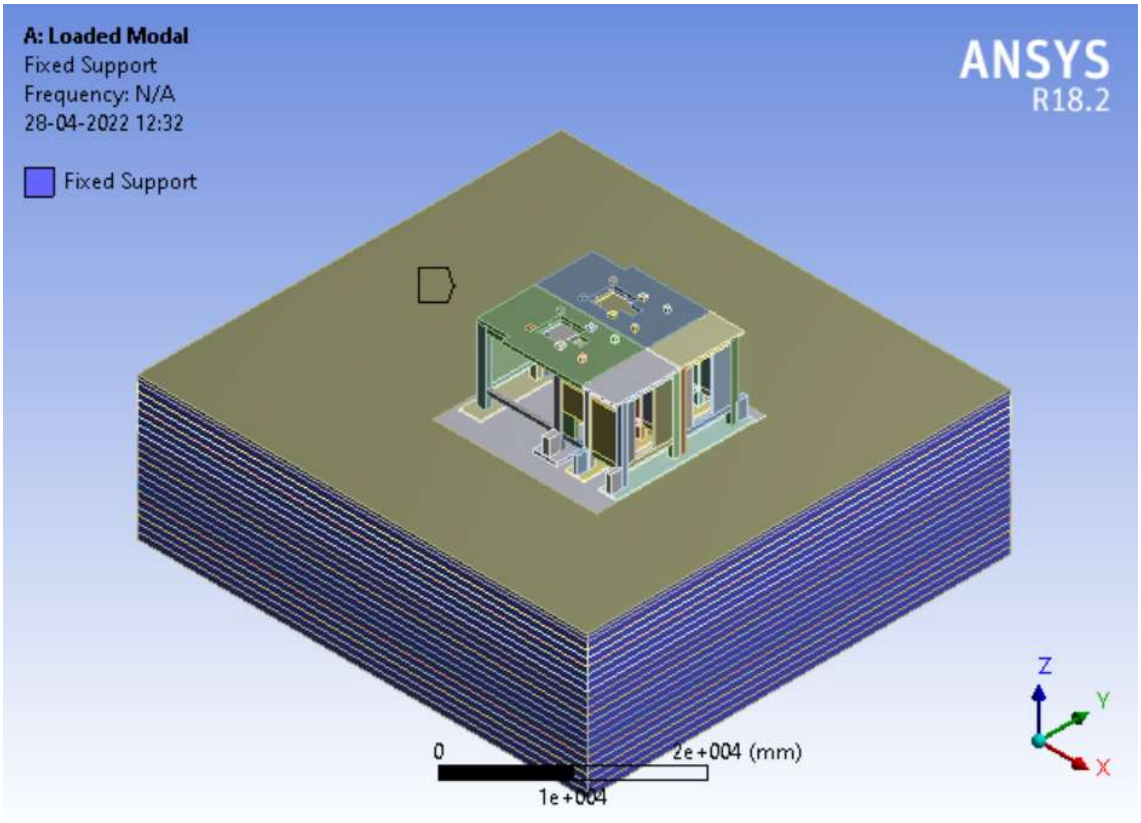


Figure 4. Boundary condition in the soil boundary

4. Analysis of the Existing Structure

Main purpose of the study is to perform harmonic analysis of the existing structure to find the root cause for the excessive structural vibration during operational condition.

4.1. Complete Model

Complete model of the existing system is shown in Figure 5a.

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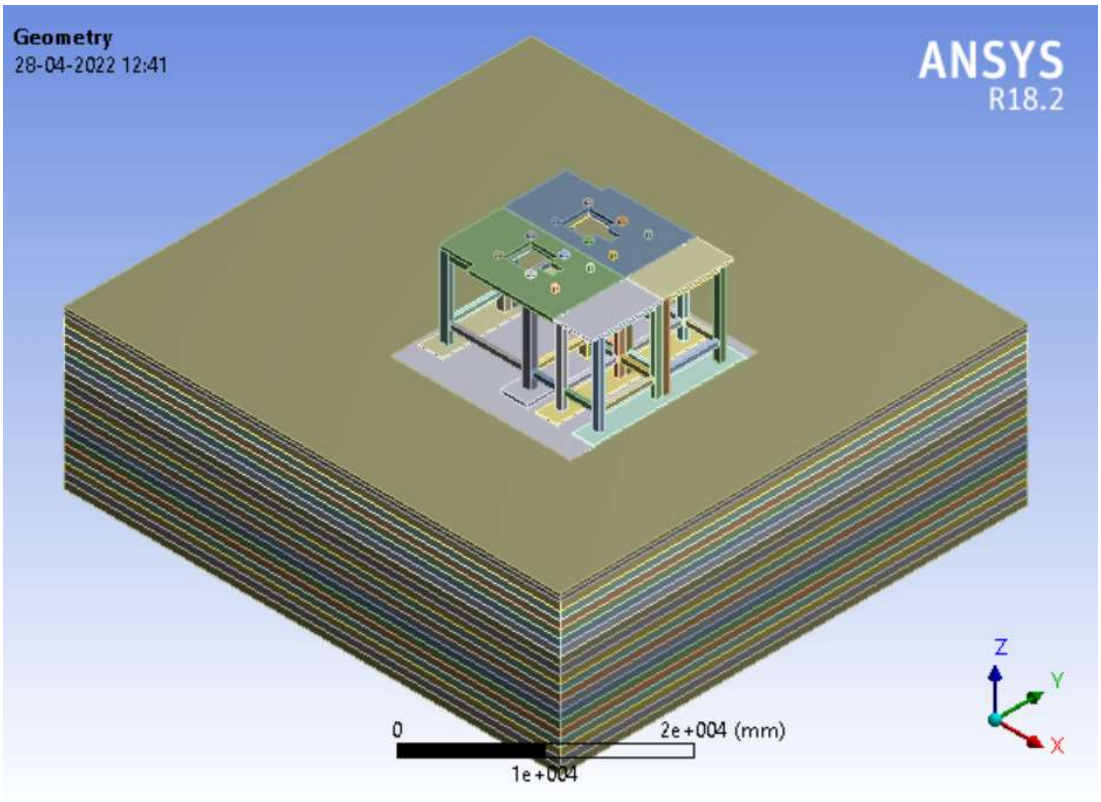


Figure 5a. Complete Model of the Tertiary Crusher system along with soil layers

4.2. Mesh

Figures 5b shows the finite element meshes of the Concrete structure & Soil layers.

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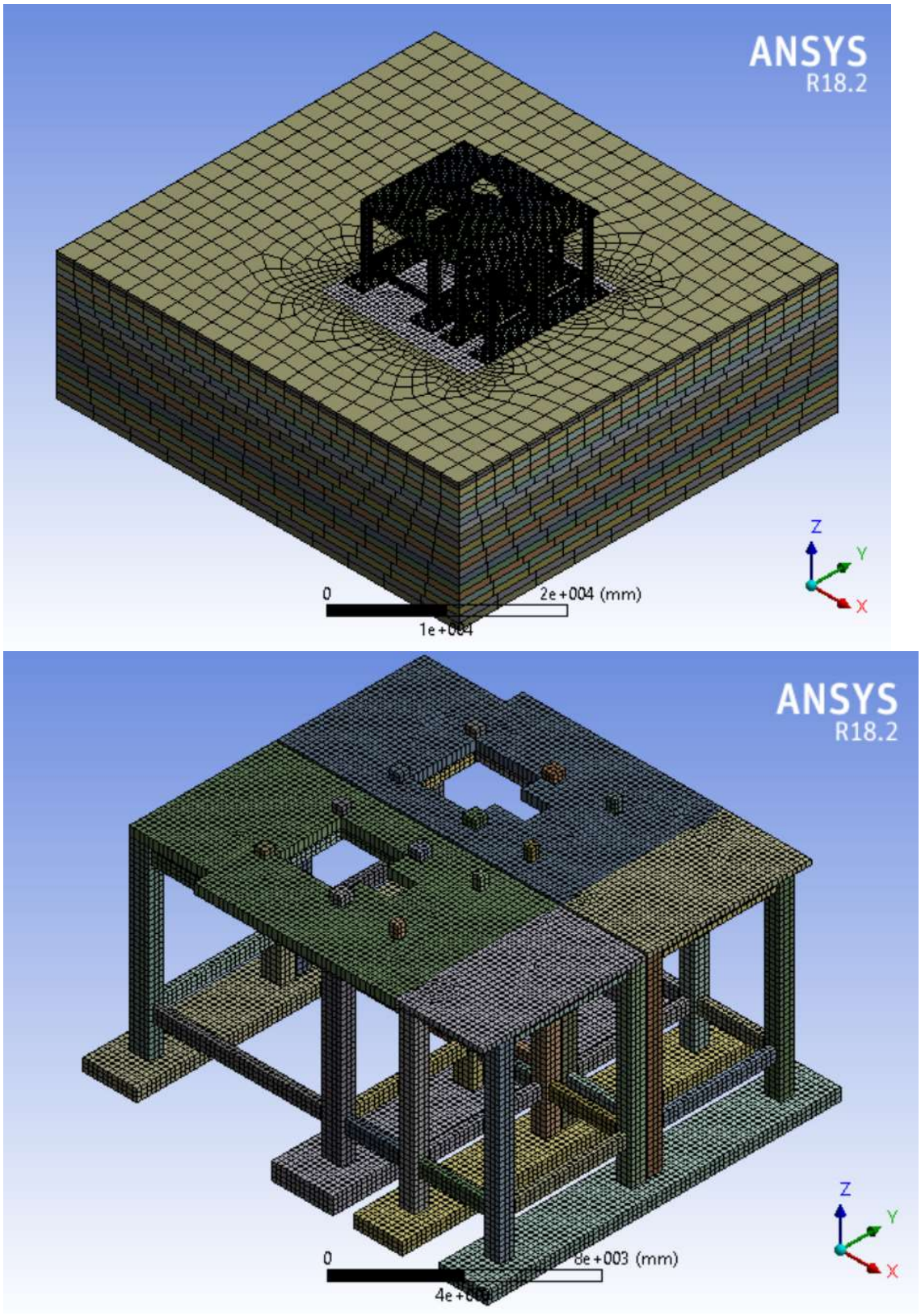


Figure 5b. Finite Element Mesh of Concrete structure

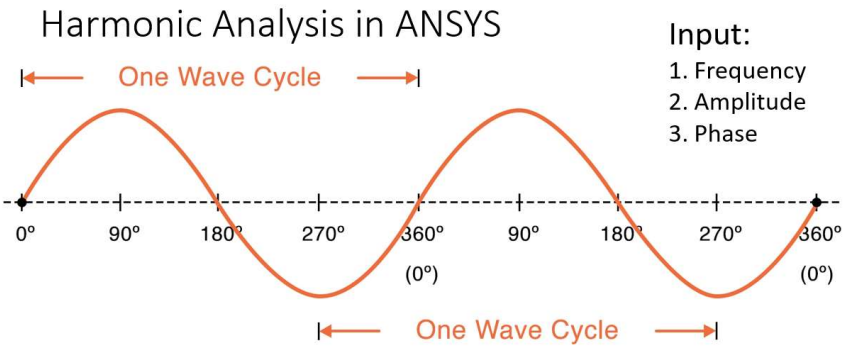
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4.3. Harmonic Analysis

Harmonic analysis is performed to determine the steady-state response of the structure subject to the dynamic crusher loads.



Harmonic analyses are **used to determine the steady- state response of a structure to loads that vary sinusoidally (harmonically) with time**.

The main purpose is to investigate if the vibration levels exceeding certain limit when subject to the dynamic load.

Harmonic analysis of the model provides the response of a system for any sinusoidal loading. Such analyses for a range of frequencies are called frequency response analysis. Plots of different quantities (displacement, velocity, etc.) vs such frequency ranges are called frequency response functions (FRF).

Harmonic analysis is done between up to 9.34 Hz frequencies with 100 intervals, shown in Table 3a.

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Details of "Analysis Settings"	
Options	
Frequency Spacing	Linear
Range Minimum	0. Hz
Range Maximum	9.23 Hz
Solution Intervals	100
User Defined Frequencies	Off
Solution Method	Mode Superposition
Include Residual Vector	No
Cluster Results	No
Modal Frequency Range	Program Controlled
Store Results At All Frequencies	Yes
Rotordynamics Controls	
Output Controls	
Damping Controls	
Constant Damping Ratio	5.e-002
Stiffness Coefficient Define By	Direct Input
Stiffness Coefficient	0.
Mass Coefficient	0.
Analysis Data Management	

Table 3a. Harmonic analysis settings

4.3.1. Results of Harmonic Analysis

Comparison between measurement & results of harmonic analysis is shown in Table 3b.

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Location Mark		Vibration (mm/sec)		
	Compare	Z	X	Y
A2 Pedestal	Measure	0.4	2.1	11.6
	Computed	0.1	4.9	14.4
A1 Pedestal	Measure	0.3	0.0	0.0
	Computed	0.2	4.9	12.3
C1 Pedestal	Measure	0.4	8.7	2.6
	Computed	0.0	5.0	10.7
B2 Pedestal	Measure	0.6	2.6	10.7
	Computed	0.1	4.7	14.4
B1 Pedestal	Measure	1.8	9.4	2.1
	Computed	0.1	4.7	12.2
C2 Pedestal	Measure	0.8	8.2	1.8
	Computed	0.0	4.8	10.7

Table 3b. Harmonic analysis results vs measurement

5. Analysis of the Proposed Structure

Main purpose of the study is to perform dynamic analysis of the proposed structure to find the structural vibration during operational condition.

5.1. Individual Component of the Model

Crusher is modelled with different components (shown in Figure-6).

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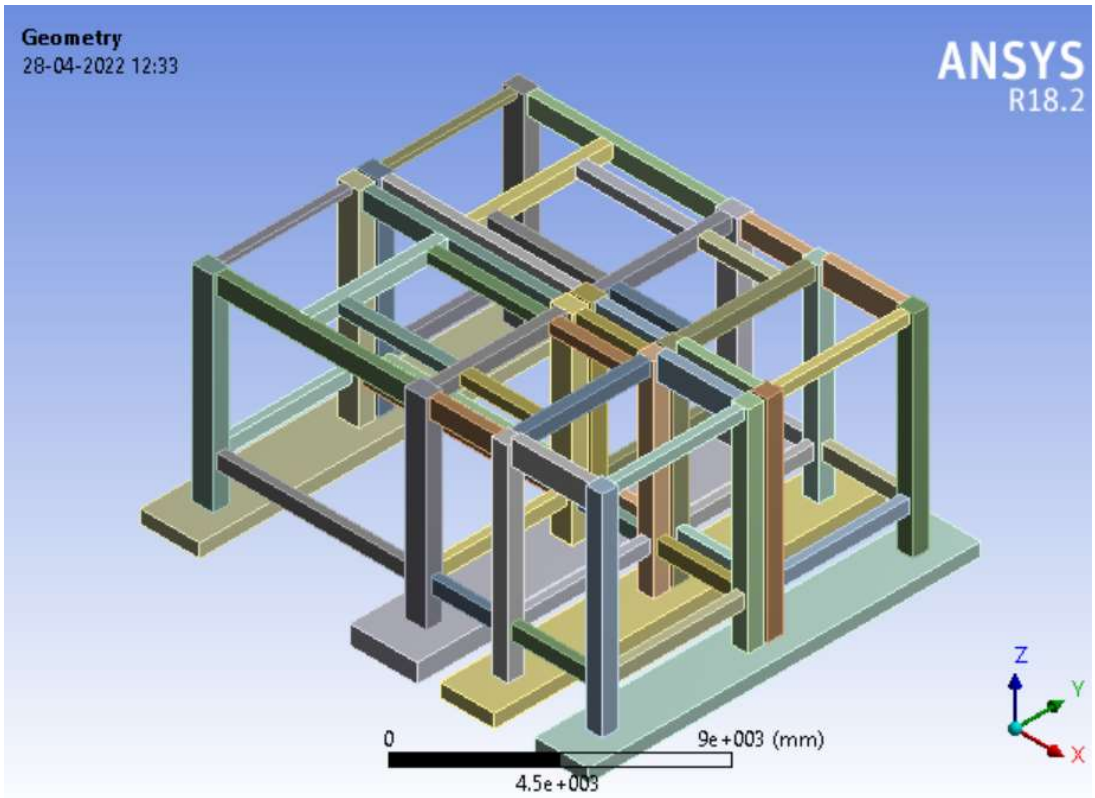
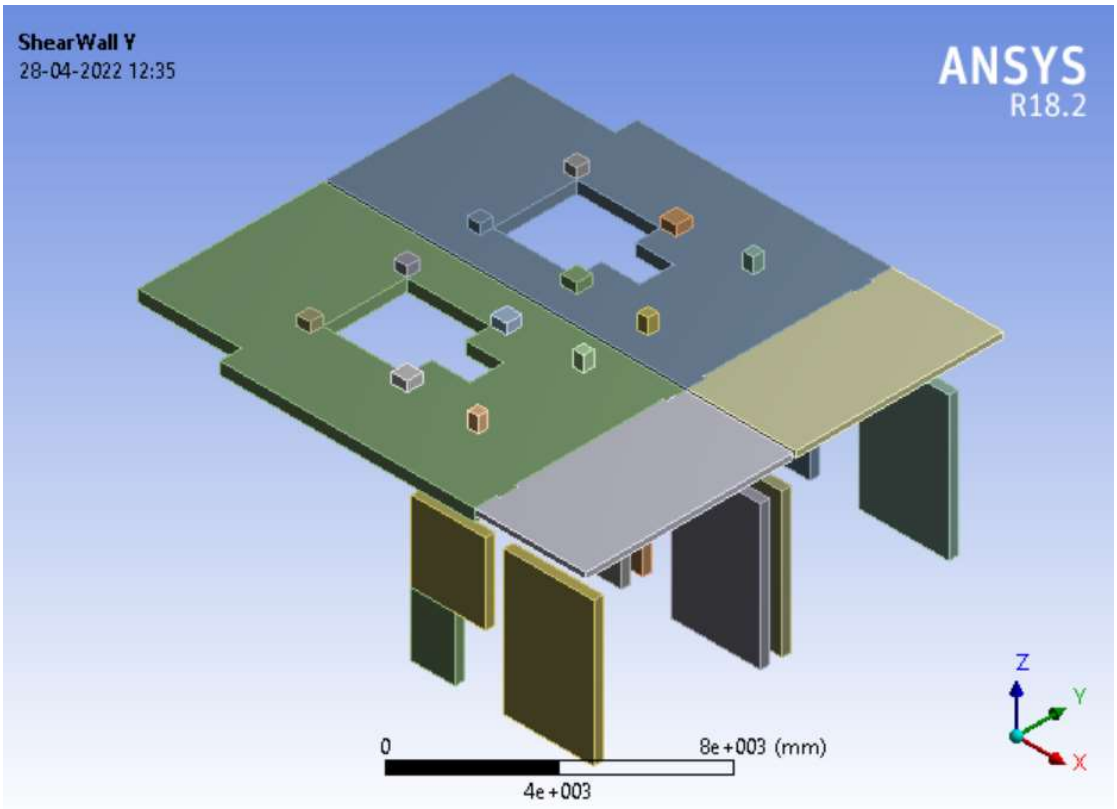


Figure 6a. Footings, beams, columns, & ties of the crusher system.

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Figure 6b. slabs, pedestals, & proposed walls in X of the crusher system.

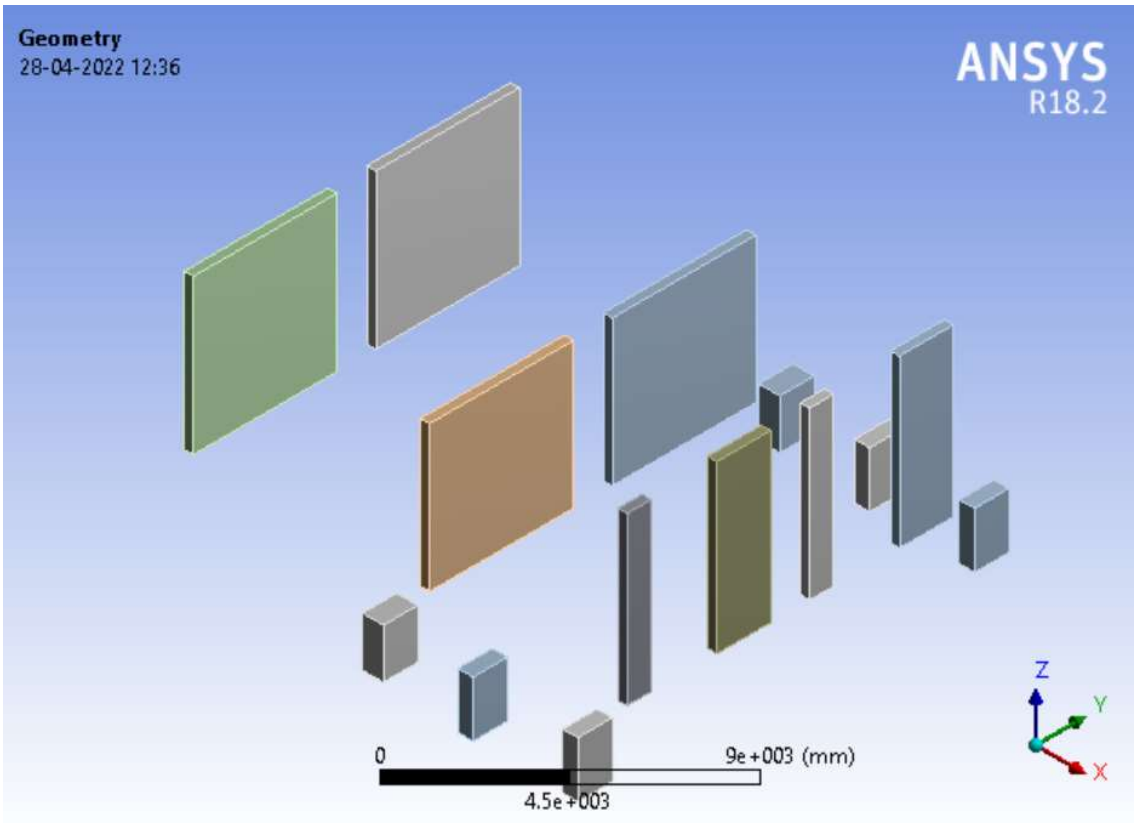


Figure 6c. proposed walls in Y and Stubs of the crusher system.

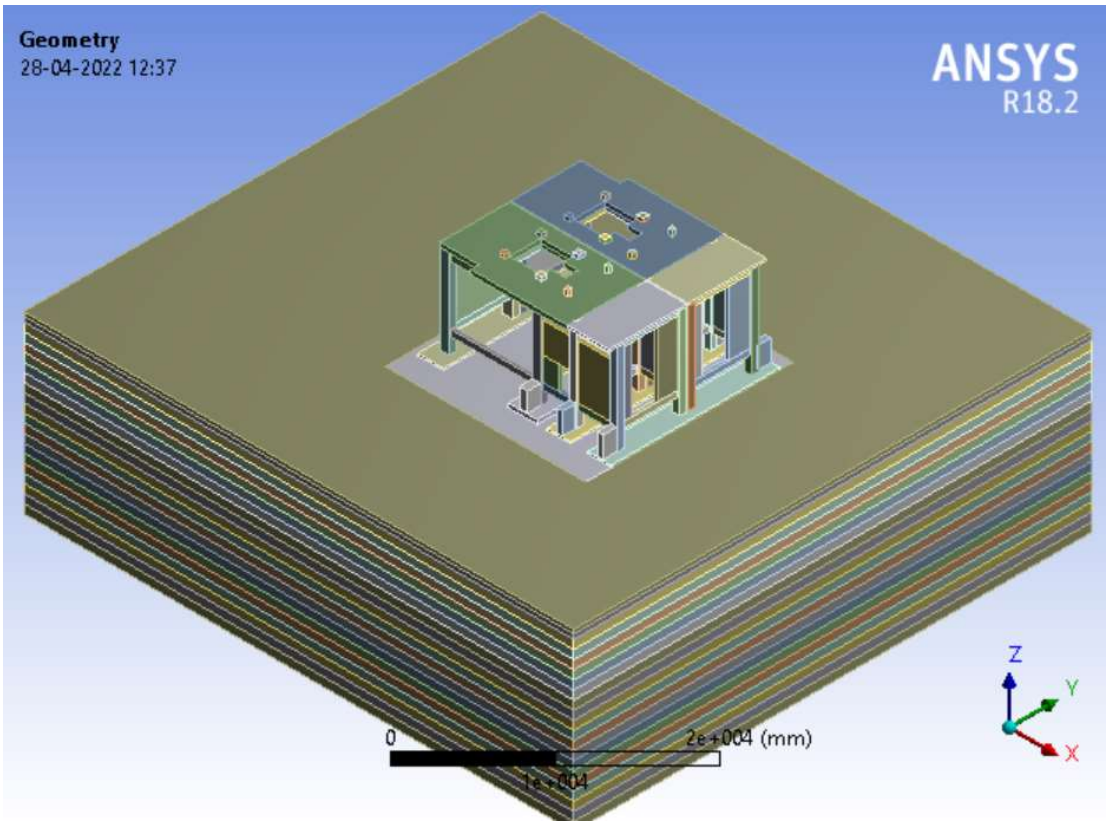
5.2. Complete Model

Complete model of the system is shown in Figure 7.

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Figure 7a. Complete Model of the Tertiary Crusher system along with soil layers

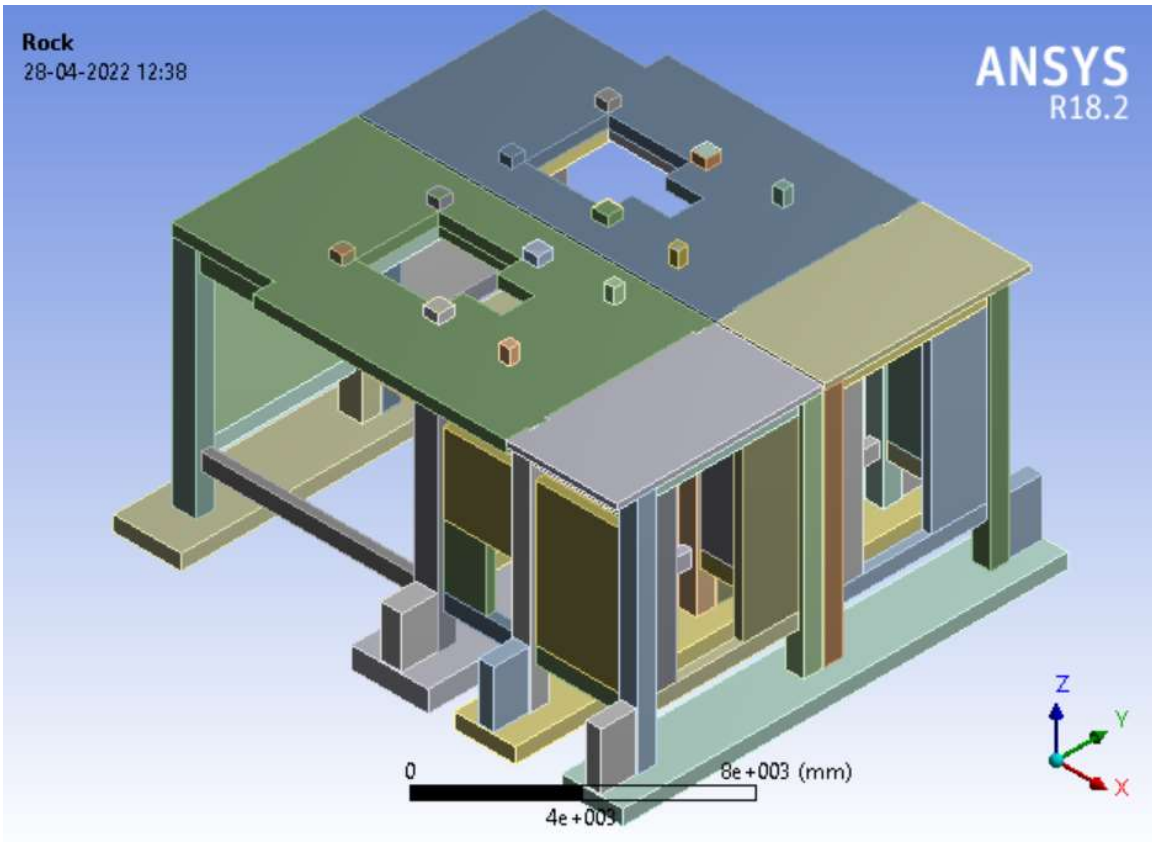


Figure 7b. Complete Model of the Tertiary Crusher system without soils

5.3. Mesh

Figures 8a & 8b shows the finite element meshes of the Concrete structure & Soil layers. The number of elements and nodes are listed in Table 4.

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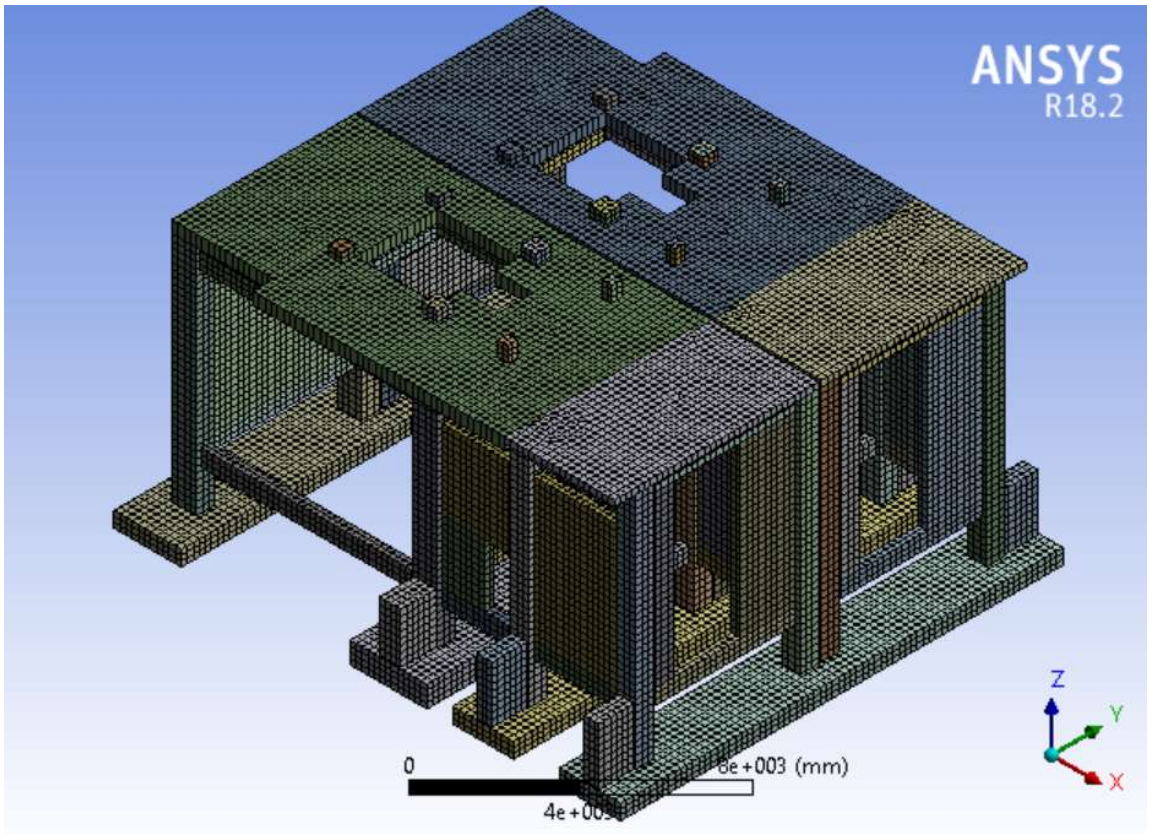


Figure 8a. Finite Element Mesh of Concrete structure

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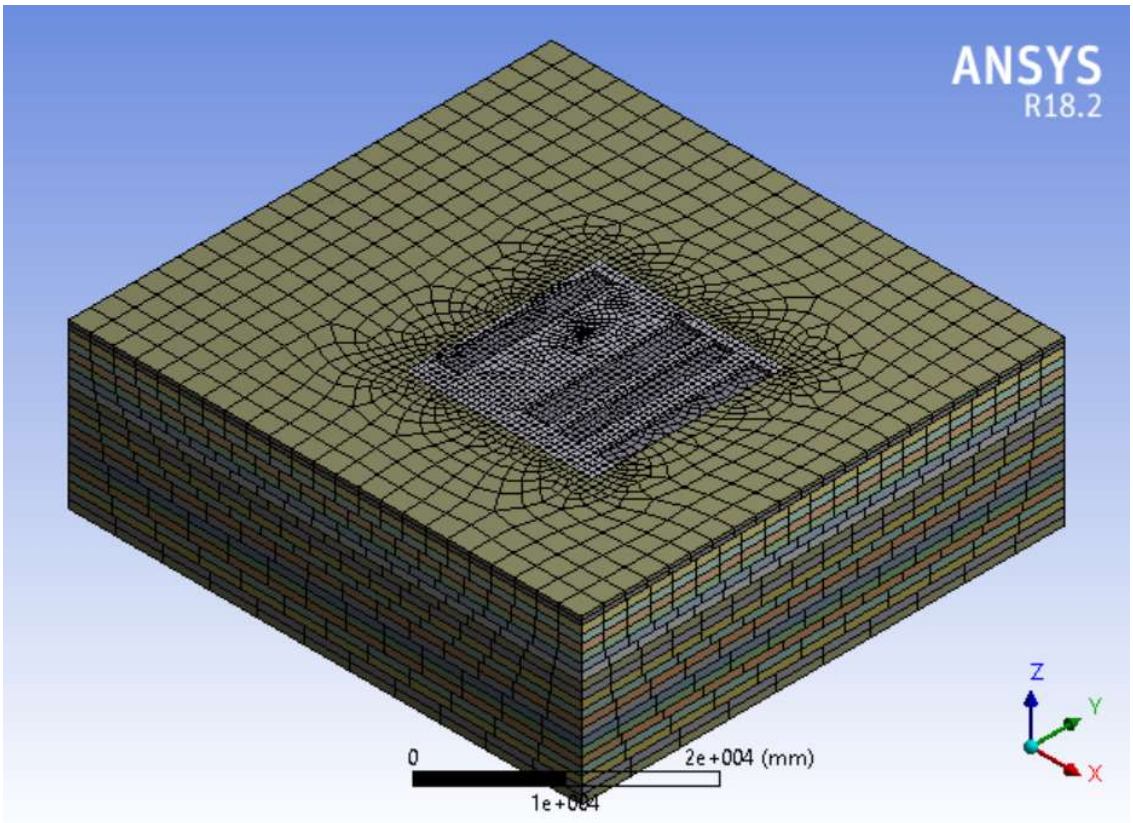


Figure 8b. Finite Element Mesh of soil layers

Statistics	
Nodes	254834
Elements	52586

Table 4. Finite Element Mesh Statistics

5.4. Modal Analysis

Table 5 and Figure-9 shows first 3 natural frequencies and first mode shapes for unloaded & loaded conditions. To verify the safe condition of the system at these frequencies, harmonic response analyses are to be performed.

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Tertiary Crusher System				
	Unloaded Condition		Loaded Condition	
Mode	Hz	Ratio	Hz	Ratio
1	9.8196	2.102698	9.3978	2.012377
2	10.435	2.234475	9.9652	2.133876
3	10.49	2.246253	10.07	2.156317

Table 5. First 12 Natural Frequencies of Tertiary Crusher with Soil layers.

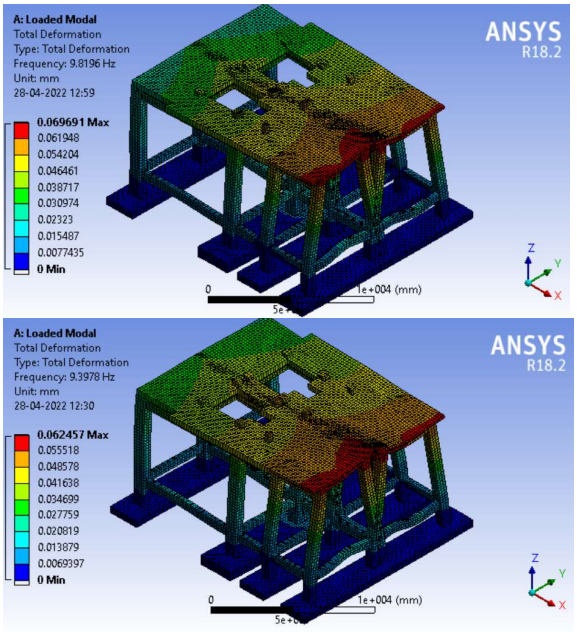


Figure 9. First Mode Shapes of Tertiary Crusher: Left: Unloaded. Right: Loaded

5.5. Harmonic Analysis

5.5.1. Relationships between different terms

Relationship between peak velocity vs peak displacement vs peak accelerations is:

Peak velocity = 2 * PI * frequency * peak displacement ……. Equation (1)

Peak acceleration = 2 * PI * frequency * peak velocity ……. Equation (2)

Where for 4.67 Hz frequency, 2 * PI * 4.67 = 29.342 rad/sec

Relationship between peak velocities vs R. M. S. velocities is:

R. M. S. velocity = peak velocity / Sqrt (2) ……. Equation (3)

5.5.2. Peak Total Velocity of Concrete Structure, Unloaded Condition: (4.67 Hz)

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Figure 11 shows the maximum peak total velocities among different phases of the Loadings

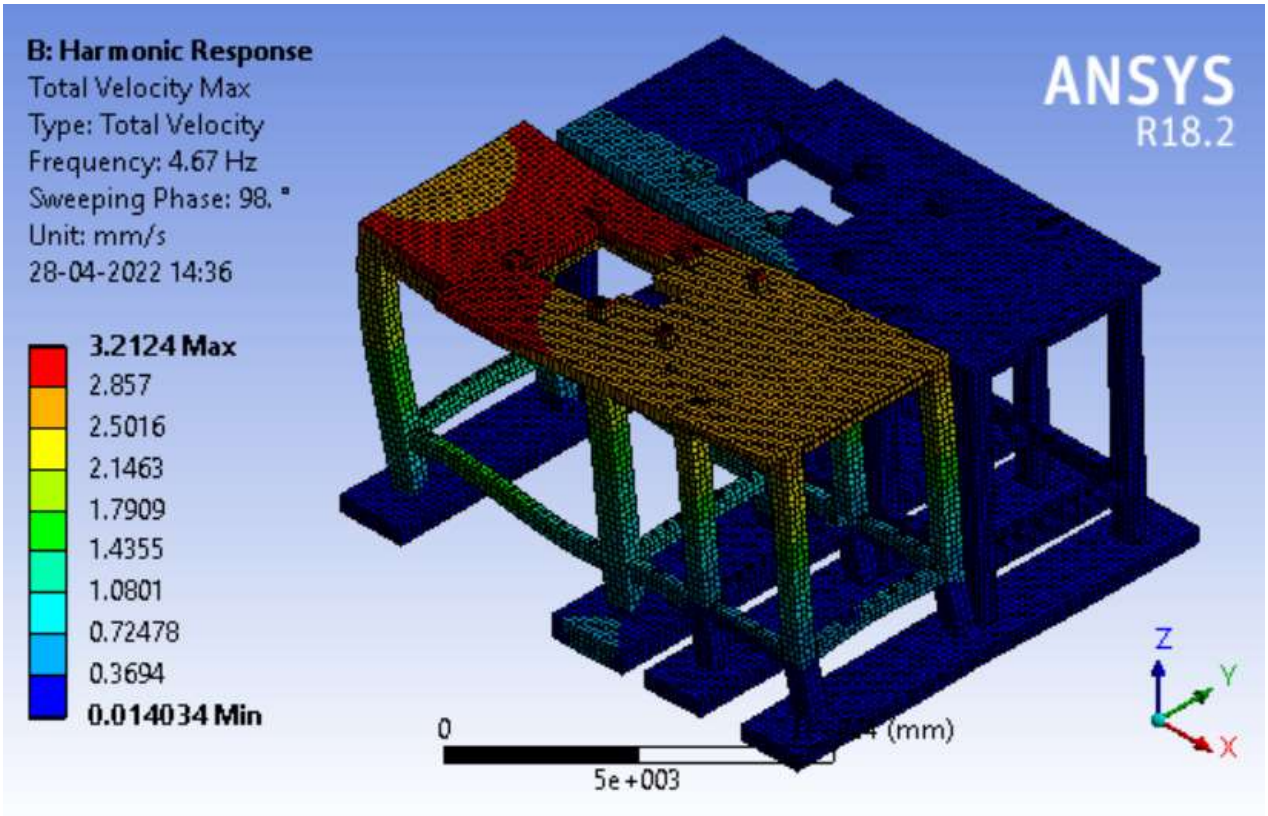


Figure 11a. maximum peak total velocity for whole structure

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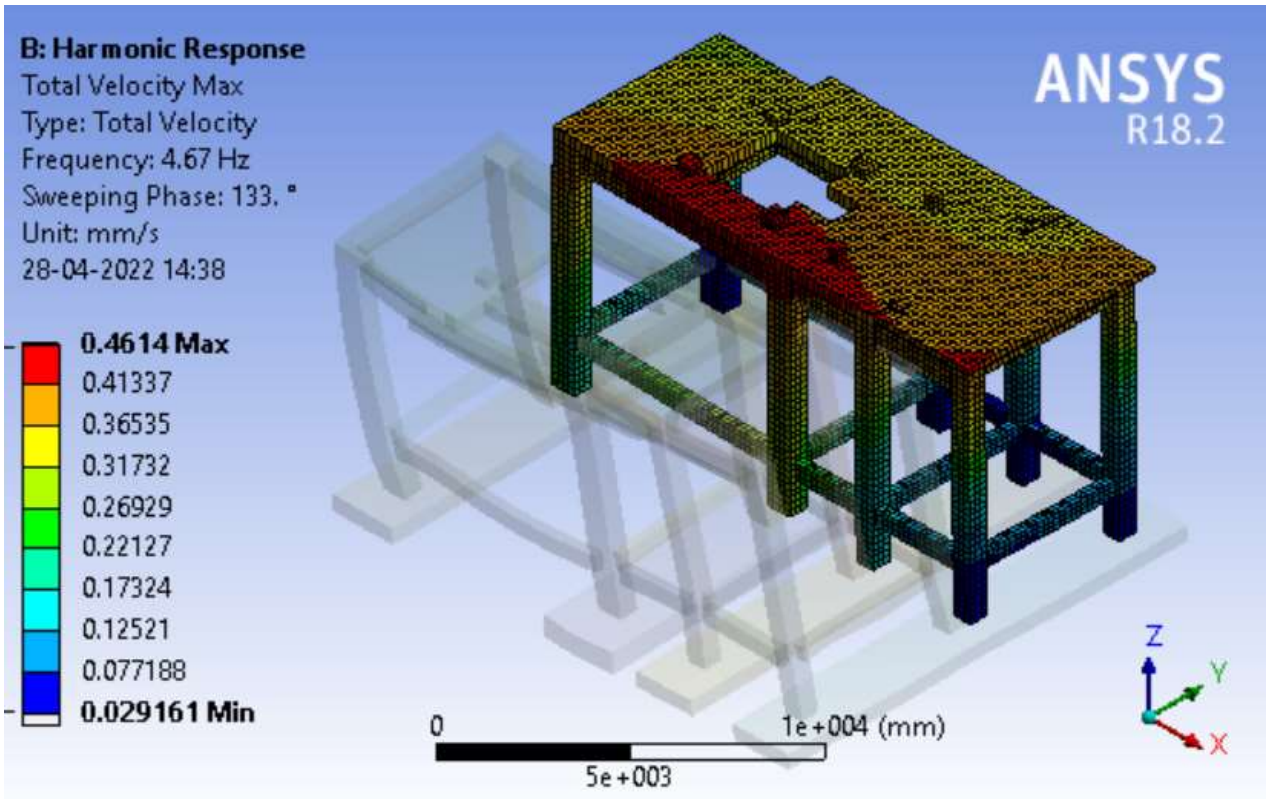


Figure 11b. maximum peak total velocity for crusher 2 structure

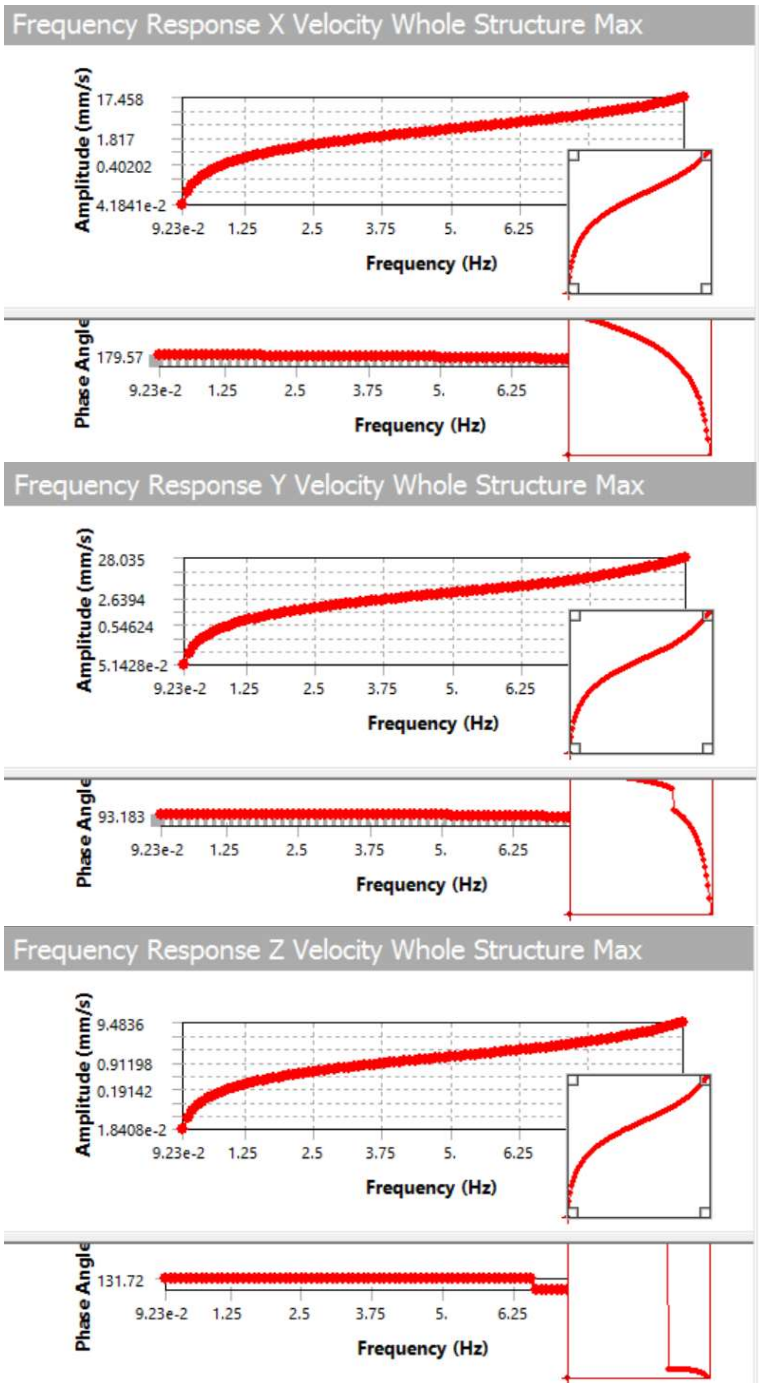
5.5.3. FRF for Peak Velocity of Concrete Structure, Unloaded Condition: (0-9.34 Hz)

Figure 12 shows the FRF of peak directional velocities for different frequencies

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	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
48	4.4304	2.4065	177.06
49	4.5227	2.4774	177.
50	4.615	2.55	176.92
51	4.7073	2.6243	176.85
52	4.7996	2.7004	176.78
53	4.8919	2.7785	176.73
54	4.9842	2.8587	176.65
55	5.0765	2.941	176.58
56	5.1688	3.0256	176.5
57	5.2611	3.1125	176.42
58	5.3534	3.202	176.34
59	5.4457	3.2941	176.25
60	5.538	3.389	176.17
61	5.6303	3.4869	176.08
62	5.7226	3.588	175.99
63	5.8149	3.6923	175.9
64	5.9072	3.8002	175.8
65	5.9995	3.9119	175.7

	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
48	4.4304	2.9562	90.12
49	4.5227	3.0435	90.022
50	4.615	3.1329	89.922
51	4.7073	3.2245	89.819
52	4.7996	3.3184	89.712
53	4.8919	3.4147	89.603
54	4.9842	3.5136	89.49
55	5.0765	3.6151	89.373
56	5.1688	3.7195	89.253
57	5.2611	3.8269	89.128
58	5.3534	3.9375	89.
59	5.4457	4.0514	88.866
60	5.538	4.1689	88.728
61	5.6303	4.2901	88.585
62	5.7226	4.4154	88.436
63	5.8149	4.5448	88.282
64	5.9072	4.6787	88.121
65	5.9995	4.8174	87.953

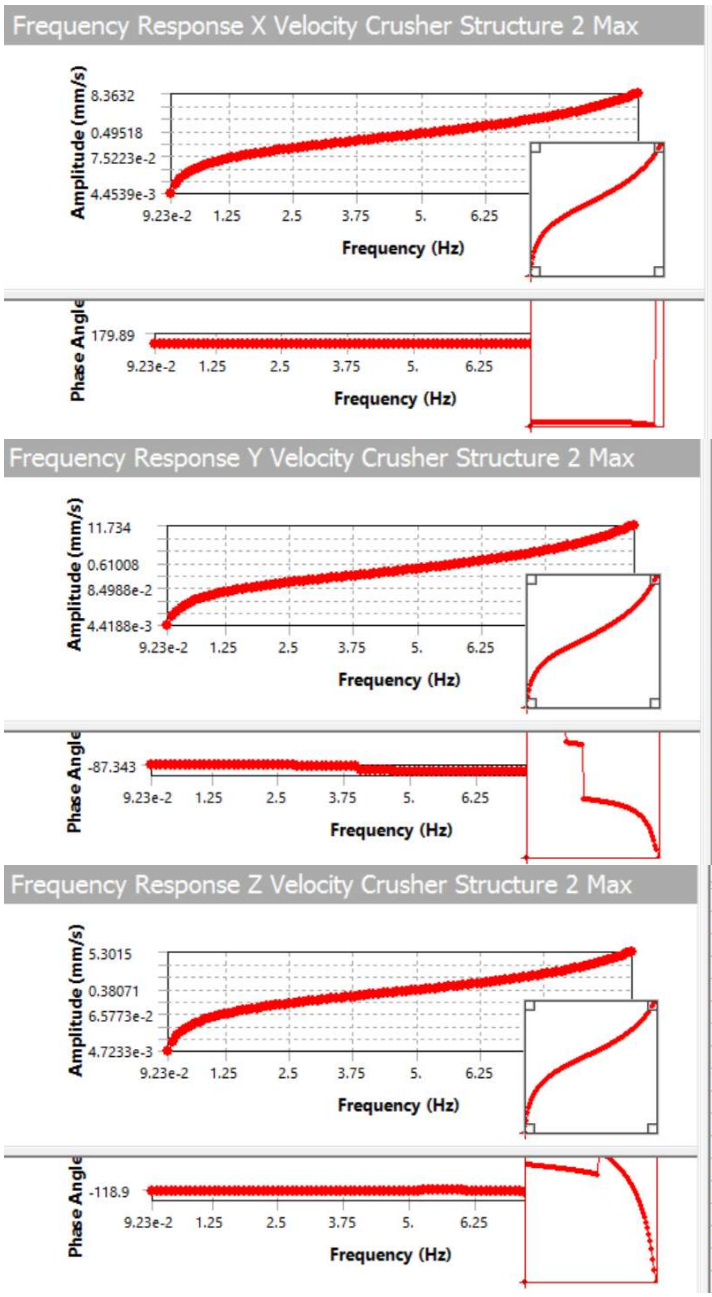
	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
48	4.4304	1.0585	128.4
49	4.5227	1.0897	128.28
50	4.615	1.1217	128.15
51	4.7073	1.1544	128.02
52	4.7996	1.188	127.89
53	4.8919	1.2224	127.76
54	4.9842	1.2578	127.62
55	5.0765	1.2941	127.47
56	5.1688	1.3314	127.32
57	5.2611	1.3697	127.17
58	5.3534	1.4092	127.01
59	5.4457	1.4498	126.84
60	5.538	1.4917	126.67
61	5.6303	1.5349	126.49
62	5.7226	1.5794	126.31
63	5.8149	1.6255	126.12
64	5.9072	1.6731	125.92
65	5.9995	1.7223	125.71

Figure 12a. FRF of peak directional velocities for whole structure

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	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
48	4.4304	0.31186	-170.69
49	4.5227	0.32399	-170.73
50	4.615	0.33662	-170.78
51	4.7073	0.34979	-170.82
52	4.7996	0.36354	-170.86
53	4.8919	0.3779	-170.91
54	4.9842	0.39292	-170.96
55	5.0765	0.40864	-171.
56	5.1688	0.42534	-169.38
57	5.2611	0.44297	-169.46
58	5.3534	0.4615	-169.54
59	5.4457	0.48099	-169.62
60	5.538	0.50151	-169.7
61	5.6303	0.52314	-169.78
62	5.7226	0.54597	-169.87
63	5.8149	0.57008	-169.96
64	5.9072	0.59558	-170.05
65	5.9995	0.6226	-170.15

	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
48	4.4304	0.2943	-136.15
49	4.5227	0.30678	-136.26
50	4.615	0.31986	-136.38
51	4.7073	0.33358	-136.49
52	4.7996	0.34798	-136.62
53	4.8919	0.36312	-136.74
54	4.9842	0.37904	-136.86
55	5.0765	0.39581	-136.99
56	5.1688	0.41349	-137.13
57	5.2611	0.43215	-137.26
58	5.3534	0.45187	-137.4
59	5.4457	0.47273	-137.54
60	5.538	0.49483	-137.69
61	5.6303	0.51826	-137.84
62	5.7226	0.54314	-138.
63	5.8149	0.5696	-138.17
64	5.9072	0.59777	-138.33
65	5.9995	0.6278	-138.51

	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
48	4.4304	0.28863	-126.8
49	4.5227	0.29804	-126.89
50	4.615	0.30774	-126.98
51	4.7073	0.31775	-127.08
52	4.7996	0.32808	-127.18
53	4.8919	0.33876	-127.28
54	4.9842	0.34981	-127.38
55	5.0765	0.36124	-127.49
56	5.1688	0.37309	-127.6
57	5.2611	0.3862	-118.9
58	5.3534	0.4003	-119.21
59	5.4457	0.41503	-119.52
60	5.538	0.43044	-119.85
61	5.6303	0.44659	-120.19
62	5.7226	0.46352	-120.53
63	5.8149	0.48131	-120.89
64	5.9072	0.50001	-121.27
65	5.9995	0.51969	-121.65

Figure 12b. FRF of peak directional velocities for crusher system 2

5.5.4. Peak Total Velocity of Concrete Structure, Loaded Condition: (4.67 Hz)

Figure 13 shows the peak total velocities among different phases of the Loadings

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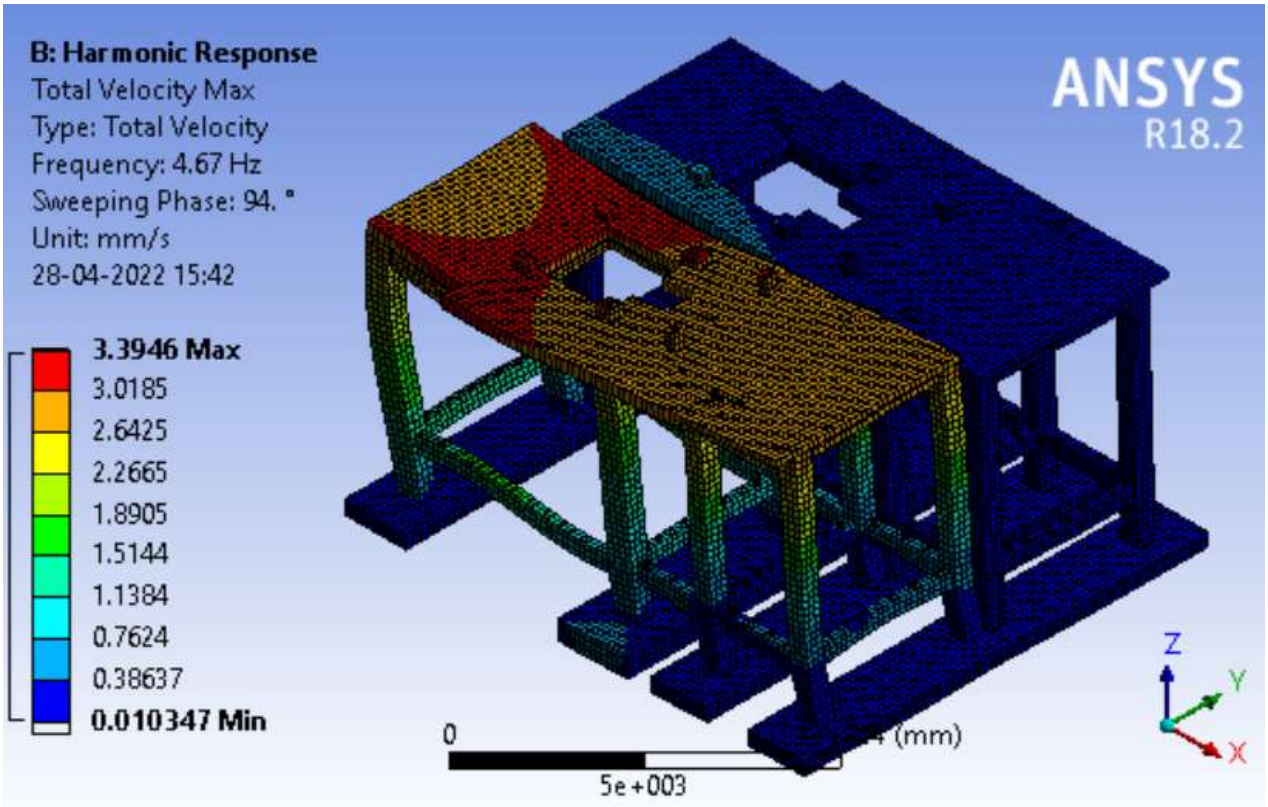
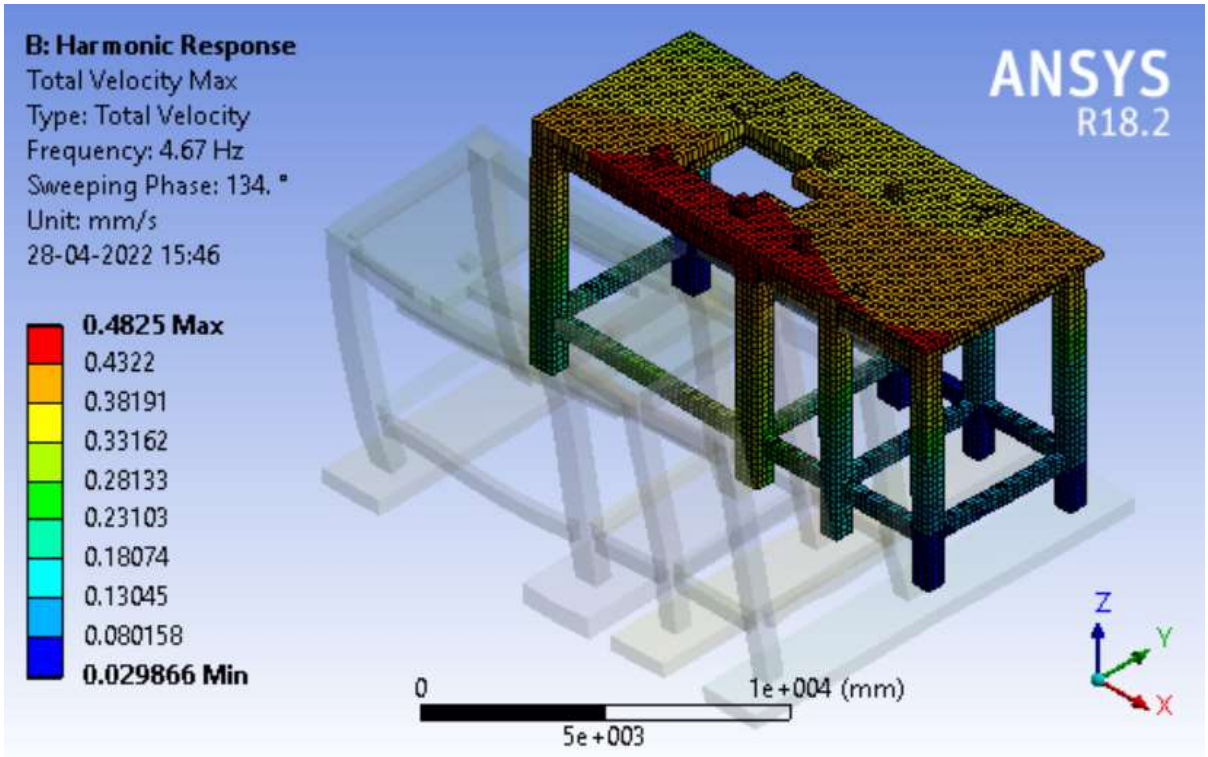


Figure 13. maximum peak total velocity for whole structure



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Figure 13b. maximum peak total velocity for crusher 2 structure

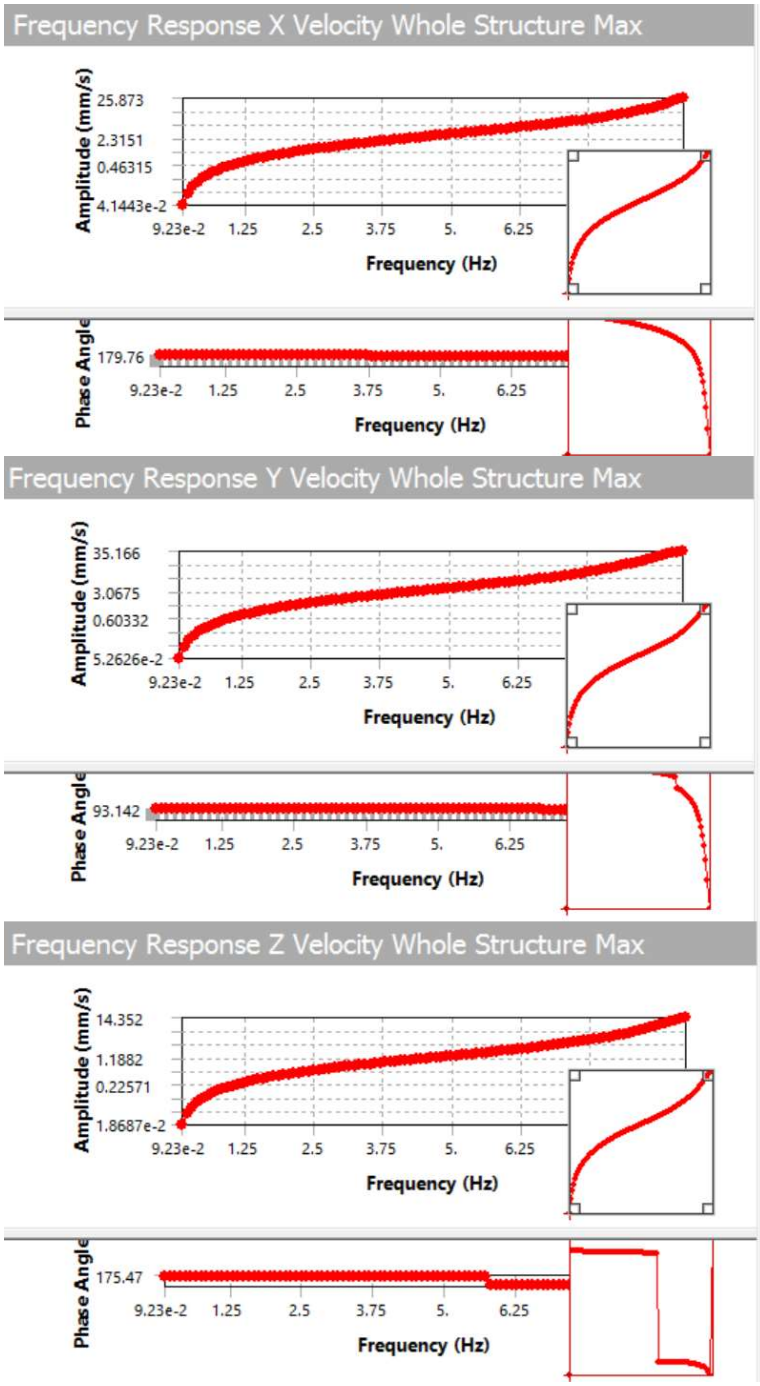
5.5.5. FRF for Peak Directional Velocity of Concrete Structure, Loaded Condition: (0-9.34 Hz)

Figure 14 shows the FRF of peak directional velocities for different frequencies

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	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
49	4.5227	2.5078	177.04
50	4.615	2.5841	176.96
51	4.7073	2.6625	176.88
52	4.7996	2.7431	176.8
53	4.8919	2.826	176.72
54	4.9842	2.9113	176.63
55	5.0765	2.9991	176.55
56	5.1688	3.0897	176.46
57	5.2611	3.1832	176.37
58	5.3534	3.2797	176.27
59	5.4457	3.3795	176.18
60	5.538	3.4827	176.08
61	5.6303	3.5895	175.97
62	5.7226	3.7003	175.87
63	5.8149	3.8152	175.76
64	5.9072	3.9346	175.65
65	5.9995	4.0587	175.53
66	6.0918	4.1878	175.41

	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
49	4.5227	3.1973	89.738
50	4.615	3.2956	89.625
51	4.7073	3.3967	89.509
52	4.7996	3.5006	89.389
53	4.8919	3.6076	89.265
54	4.9842	3.7179	89.136
55	5.0765	3.8316	89.003
56	5.1688	3.9489	88.865
57	5.2611	4.0702	88.721
58	5.3534	4.1955	88.572
59	5.4457	4.3252	88.417
60	5.538	4.4596	88.255
61	5.6303	4.5989	88.087
62	5.7226	4.7435	87.911
63	5.8149	4.8937	87.727
64	5.9072	5.05	87.535
65	5.9995	5.2127	87.334
66	6.0918	5.3824	87.122

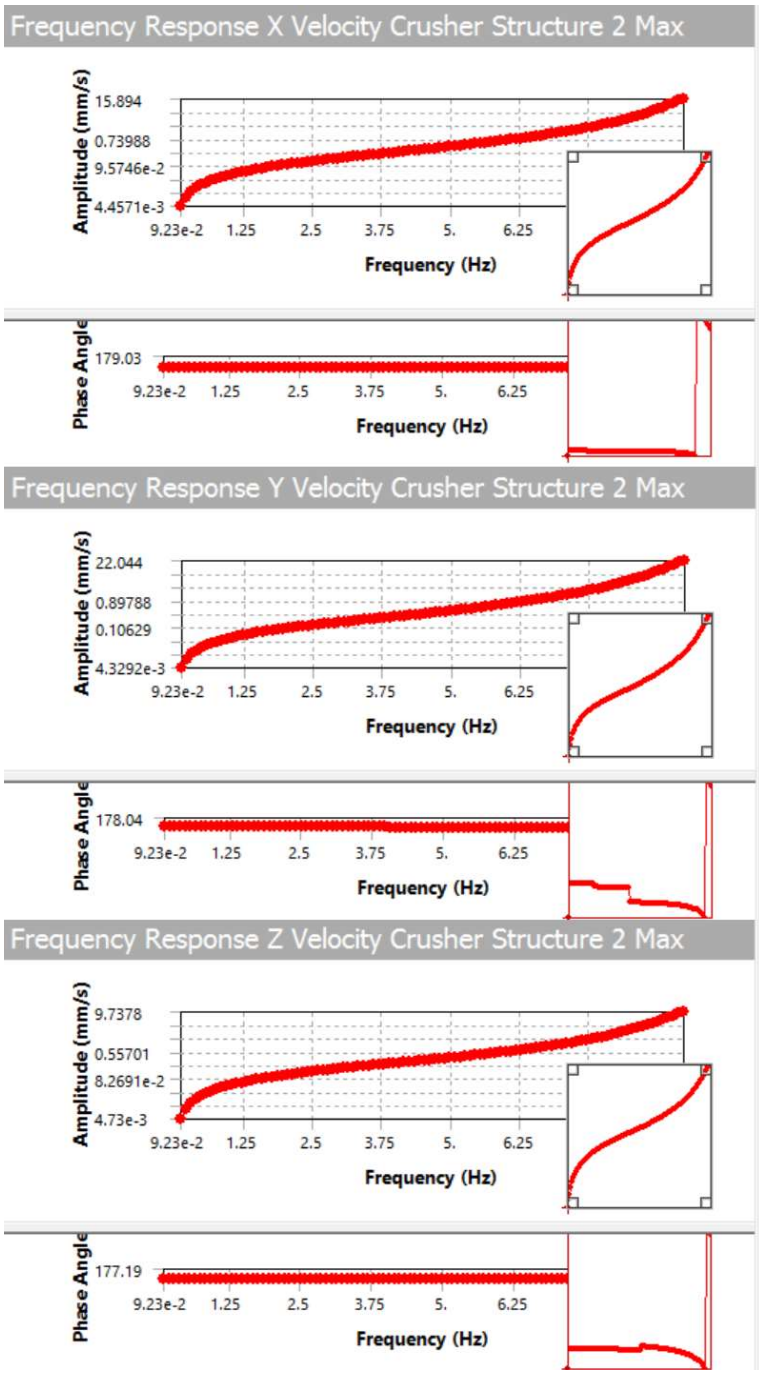
	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
49	4.5227	1.1354	126.58
50	4.615	1.1702	126.45
51	4.7073	1.206	126.31
52	4.7996	1.2429	126.17
53	4.8919	1.2808	126.02
54	4.9842	1.3198	125.86
55	5.0765	1.36	125.71
56	5.1688	1.4015	125.54
57	5.2611	1.4443	125.37
58	5.3534	1.4885	125.19
59	5.4457	1.5343	125.01
60	5.538	1.5816	124.82
61	5.6303	1.6306	124.62
62	5.7226	1.6815	124.41
63	5.8149	1.7352	-142.18
64	5.9072	1.7942	-142.24
65	5.9995	1.856	-142.3
66	6.0918	1.9207	-142.36

Figure 14a. FRF of peak directional velocities for total structure

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	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
49	4.5227	0.33718	-170.89
50	4.615	0.35104	-170.95
51	4.7073	0.36555	-171.01
52	4.7996	0.38094	-169.34
53	4.8919	0.39723	-169.42
54	4.9842	0.41435	-169.5
55	5.0765	0.43237	-169.59
56	5.1688	0.45136	-169.67
57	5.2611	0.47139	-169.77
58	5.3534	0.49254	-169.86
59	5.4457	0.5149	-169.96
60	5.538	0.53857	-170.06
61	5.6303	0.56367	-170.17
62	5.7226	0.5903	-170.28
63	5.8149	0.61861	-170.39
64	5.9072	0.64874	-170.51
65	5.9995	0.68087	-170.63
66	6.0918	0.71517	-170.76

	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
49	4.5227	0.31895	-136.39
50	4.615	0.33439	-136.52
51	4.7073	0.34961	-136.64
52	4.7996	0.36566	-136.77
53	4.8919	0.38261	-136.91
54	4.9842	0.40053	-137.04
55	5.0765	0.41949	-137.18
56	5.1688	0.43958	-137.32
57	5.2611	0.46091	-137.47
58	5.3534	0.48356	-137.62
59	5.4457	0.50766	-137.78
60	5.538	0.53335	-137.95
61	5.6303	0.56075	-138.12
62	5.7226	0.59004	-138.29
63	5.8149	0.62139	-138.48
64	5.9072	0.655	-138.67
65	5.9995	0.69109	-138.87
66	6.0918	0.72993	-139.08

	Frequency [Hz]	Amplitude [mm/s]	Phase Angle [°]
49	4.5227	0.30722	-127.05
50	4.615	0.3177	-127.15
51	4.7073	0.32856	-127.26
52	4.7996	0.33981	-127.37
53	4.8919	0.35163	-118.03
54	4.9842	0.36513	-118.34
55	5.0765	0.37926	-118.67
56	5.1688	0.39406	-119.
57	5.2611	0.40959	-119.35
58	5.3534	0.42589	-119.7
59	5.4457	0.44304	-120.07
60	5.538	0.4611	-120.46
61	5.6303	0.48015	-120.85
62	5.7226	0.50026	-121.26
63	5.8149	0.52153	-121.69
64	5.9072	0.54406	-122.13
65	5.9995	0.56797	-122.59
66	6.0918	0.59337	-123.07

Figure 14b. FRF of peak directional velocities crusher system 2

5.5.6. Summary of Results

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Table 6a shows the maximum Peak displacement, Velocity, and acceleration in the concrete structure for unloaded & loaded conditions. Table 6b shows the R.M.S. velocity of the structure.

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Only Crusher System 1 is Running					
Total Structure		Total	X	Y	Z
		Peak	Peak	Peak	Peak
Maximum Velocities (mm/sec)	Unloaded	3.212	2.600	3.150	1.130
	Loaded	3.395	2.620	3.330	1.190
Maximum Displacement (micron)	Unloaded	109.480	88.609	107.353	38.511
	Loaded	115.689	89.290	113.487	40.556
Maximum Acceleration (m/sec2)	Unloaded	0.094	0.076	0.092	0.033
	Loaded	0.100	0.077	0.098	0.035

Only Crusher System 1 is Running					
Crusher 2 Structure		Total	X	Y	Z
		Peak	Peak	Peak	Peak
Maximum Velocities (mm/sec)	Unloaded	0.461	0.340	0.330	0.300
	Loaded	0.483	0.360	0.340	0.320
Maximum Displacement (micron)	Unloaded	15.725	11.587	11.246	10.224
	Loaded	16.444	12.269	11.587	10.906
Maximum Acceleration (m/sec2)	Unloaded	0.014	0.010	0.010	0.009
	Loaded	0.014	0.011	0.010	0.009

Both Crusher Systems are Running (Considering Absolute Superposition)					
Total Structure		Total	X	Y	Z
		Peak	Peak	Peak	Peak
Maximum Velocities (mm/sec)	Unloaded	3.674	2.940	3.480	1.430
	Loaded	3.877	2.980	3.670	1.510
Maximum Displacement (micron)	Unloaded	125.204	100.196	118.599	48.735
	Loaded	132.133	101.559	125.075	51.461
Maximum Acceleration (m/sec2)	Unloaded	0.108	0.086	0.102	0.042
	Loaded	0.114	0.087	0.108	0.044

Table 6a. Peak Displacement, Velocity, & Acceleration

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Only Crusher System 1 is Running					
Total Structure		Total	X	Y	Z
		RMS	RMS	RMS	RMS
Maximum Velocities (mm/sec)	Unloaded	2.272	1.838	2.227	0.799
	Loaded	2.400	1.853	2.355	0.841

Only Crusher System 1 is Running					
Crusher 2 Structure		Total	X	Y	Z
		RMS	RMS	RMS	RMS
Maximum Velocities (mm/sec)	Unloaded	0.326	0.240	0.233	0.212
	Loaded	0.341	0.255	0.240	0.226

Both Crusher Systems are Running (Considering Absolute Superposition)					
Total Structure		Total	X	Y	Z
		RMS	RMS	RMS	RMS
Maximum Velocities (mm/sec)	Unloaded	2.598	2.079	2.461	1.011
	Loaded	2.742	2.107	2.595	1.068

Table 6b. RMS Velocity of the structure

6. Conclusions

The maximum peak vibration level for the x, y and z directions over the excitation forcing frequency 4.67 Hz is computed in the finite element model. The absolute maximum R.M.S. velocity level computed for frequency is 2.598 & 2.742 mm/s for unloaded & loaded condition respectively. Peak displacements computed 125 & 132 microns for unloaded & loaded conditions respectively.

7. References

[1] ANSYS Mechanical Software

[2] MANSOURAH - MASSARAH GOLD PROJECT GENERAL ARRANGEMENT TERTIARY CRUSHING TERTIARY CONE CRUSHER 3120-CRH-001 (MD - 216 -3130 - EG- ME -GAD -V500-1001) Rev -C01.

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8. Appendix 1

PROPRIETARY & CONFIDENTIAL

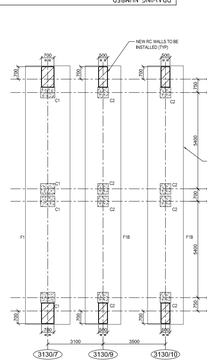
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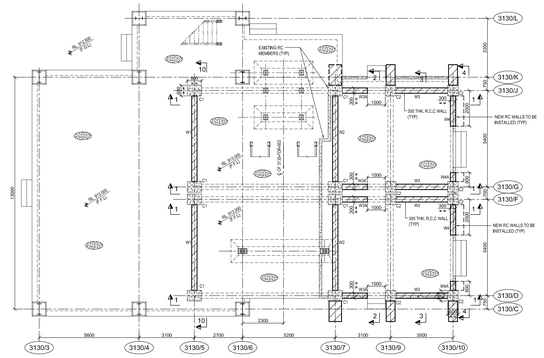
REV: A02

DRAWING NUMBER: MD - 216 - 3130 - EG - CV - GAD - 1009

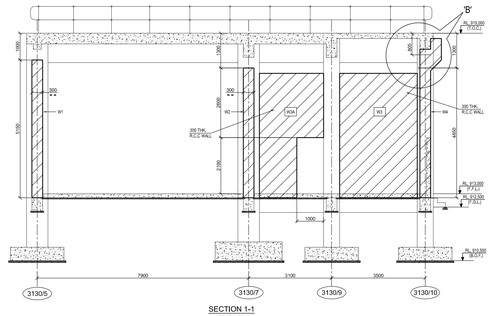




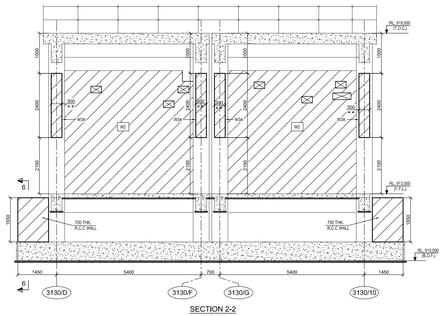
PART LAYOUT OF FOUNDATION AT RL. 910.500 (B.O.F) AND NEW RC WALLS
(SCALE - 1:75)



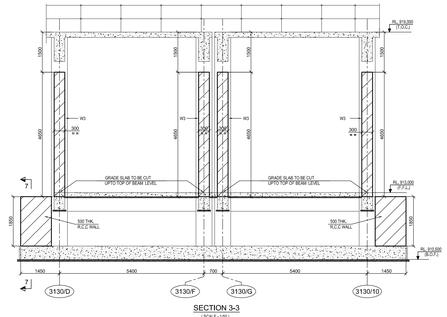
PLAN OF GRADE SLAB AT RL.913.000 (F.F.L), TIE BEAM AT RL. 912.850 (T.O.B) AND NEW RC WALL
(SCALE - 1:75)



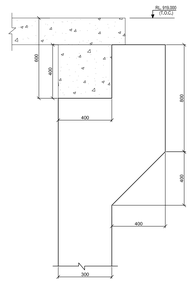
SECTION 1-1
(SCALE - 1:50)



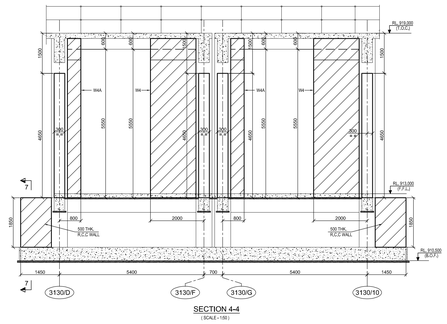
SECTION 2-2
(SCALE - 1:50)



SECTION 3-3
(SCALE - 1:50)

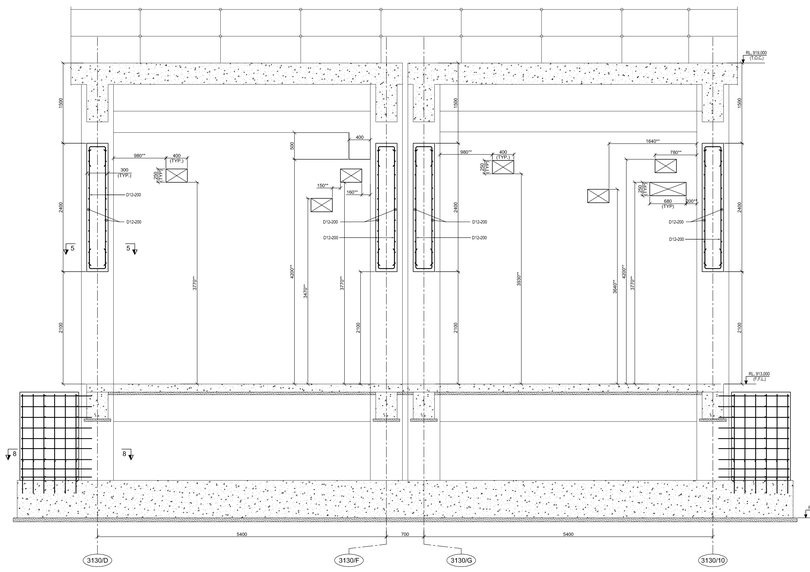


DETAIL- B
(SCALE-1:75)

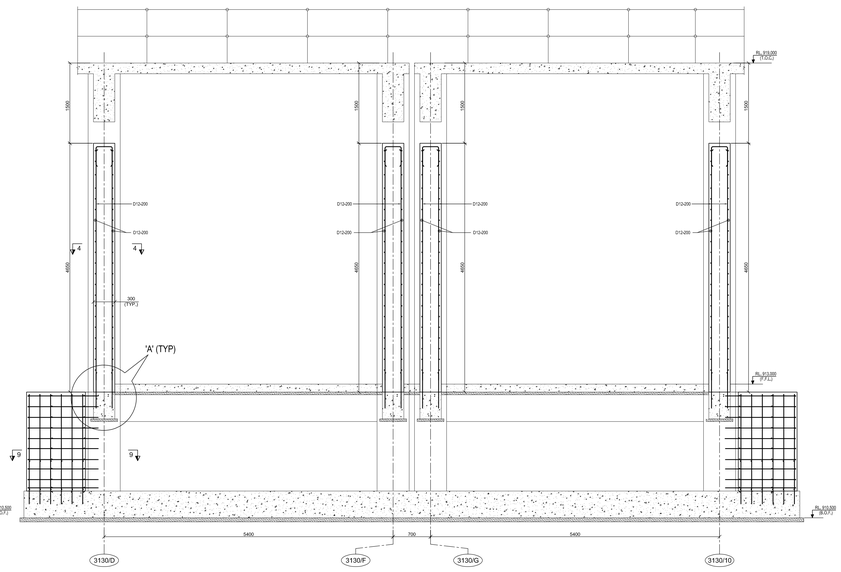


SECTION 4-4
(SCALE - 1:50)

** DIMENSIONS SHALL BE AS PER ACTUAL SITE MEASUREMENT

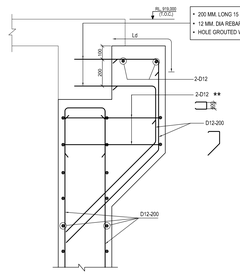


R.C.C DETAIL OF SECTION 2-2
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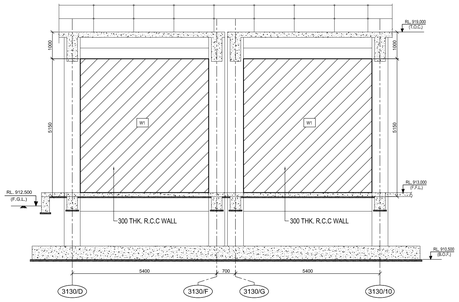


R.C.C DETAIL OF SECTION 3-3
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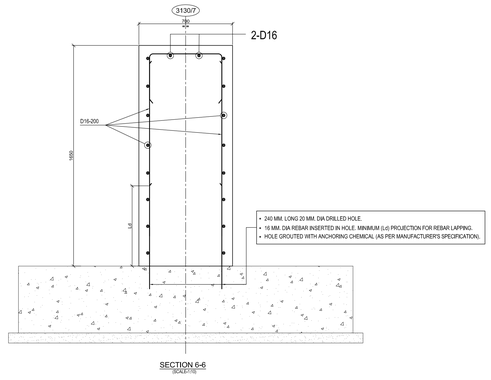
- 200 MM. LONG 15 MM. DIA DRILLED HOLE.
- 12 MM. DIA REBAR INSERTED IN HOLE, MINIMUM (Ld) PROJECTION FOR REBAR LAPPING.
- HOLE GROUTED WITH ANCHORING CHEMICAL (AS PER MANUFACTURER'S SPECIFICATION).



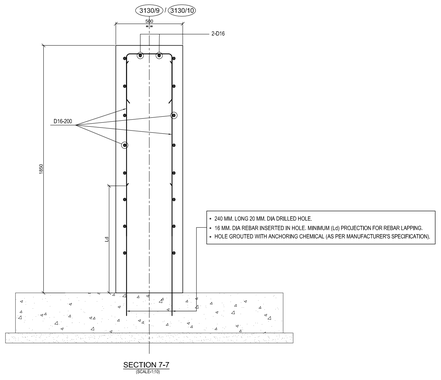
DETAIL-B
(R.C.C DETAILS)
(2 NOS FOR AA & B NOS FOR A)
(SCALE-1:15)



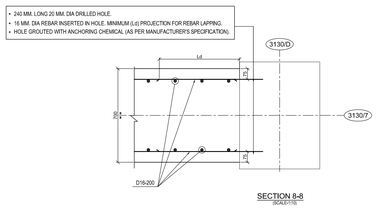
SECTION 10-10
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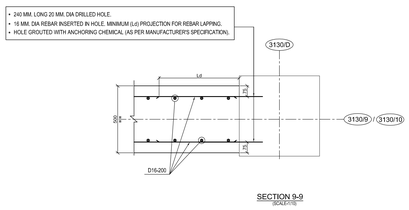
SECTION 6-6
(SCALE-1:10)



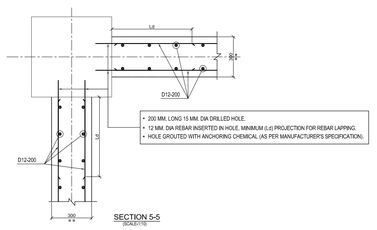
SECTION 7-7
(SCALE-1:10)



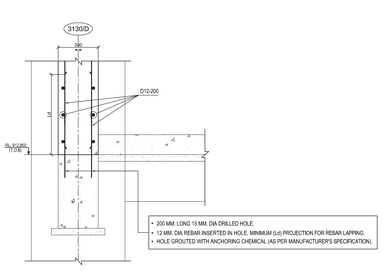
SECTION 8-8
(SCALE-1:10)



SECTION 9-9
(SCALE-1:10)



SECTION 5-5
(SCALE-1:10)



DETAIL-A
(SCALE-1:10)

WALL MARK	WALL TOP LEVEL
W1	RL 918.000
W2	RL 917.700
W3	RL 917.500
W3A	RL 917.500
W4	RL 918.400
W4A	RL 918.400

LEGENDS :-
RL.- REDUCED LEVEL
T.O.C. - TOP OF CONCRETE
F.G.L. - FINISHED GROUND LEVEL
F.F.L. - FINISHED FLOOR LEVEL
B.O.F. - BOTTOM OF FOUNDATION
T.O.B. - TOP OF BEAM
R.C.C - REINFORCED CEMENT CONCRETE
NEW WALL CONSTRUCTION

NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES.
2. GRADE OF CONCRETE SHALL BE C35 FOR WALL.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 – GRADE 60 (fy = 420 MPa).
4. QUICK SETTING ADMIXTURE SHALL BE USED FOR REDUCING THE SETTING TIME OF THE CONCRETE.
5. CLEAR COVER TO MAIN REINFORCEMENT FOR WALL SHALL BE 40 MM ABOVE F.G.L & 75 MM BELOW F.G.L
6. MINIMUM LAP LENGTH OF REINFORCEMENT SHALL BE 40 X DIA. OF BAR.
7. DO NOT SCALE, PLEASE ASK.

		A03	15/10/2022	ISSUED OF APPROVAL	HSK	SBY	RPK	AG															
MD-216-3130-EG-CV-GAD-1008	TERTIARY CRUSHING DETAILS & SECTIONS	A02	29/04/2022	ISSUED OF APPROVAL	SYA	SBY	RPK	AG															
MD-216-3130-EG-CV-GAD-1007	TERTIARY CRUSHING LAYOUT & DETAIL	A01	12/04/2022	ISSUED OF APPROVAL	SYA	SBY	RPK	AG															
DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP	DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP				

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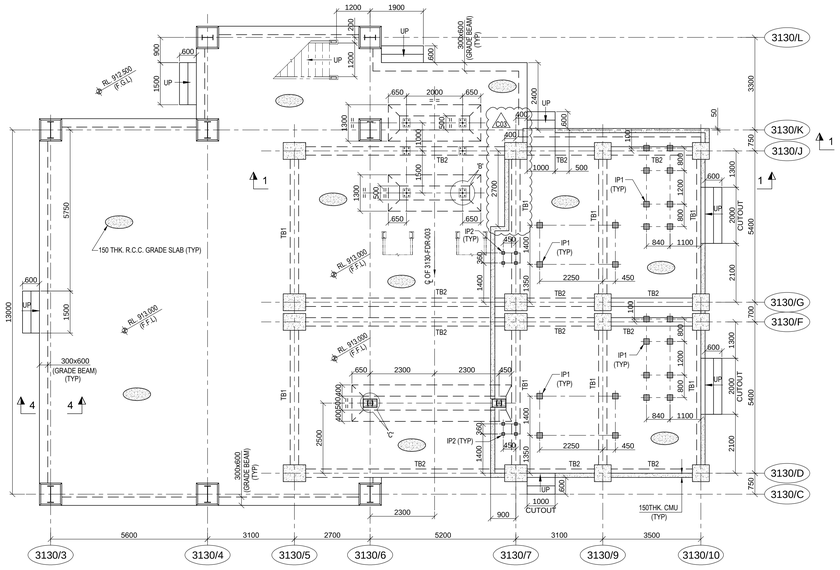
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CHECKED	SBY	11/04/2022
ENGINEER	RPK	08/04/2022
APPROVED	AG	12/04/2022
EMPLOYER		
SCALE	1:75,1:50,1:25,1:10	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
MODIFICATION OF CRUSHER SUPPORT STRUCTURE

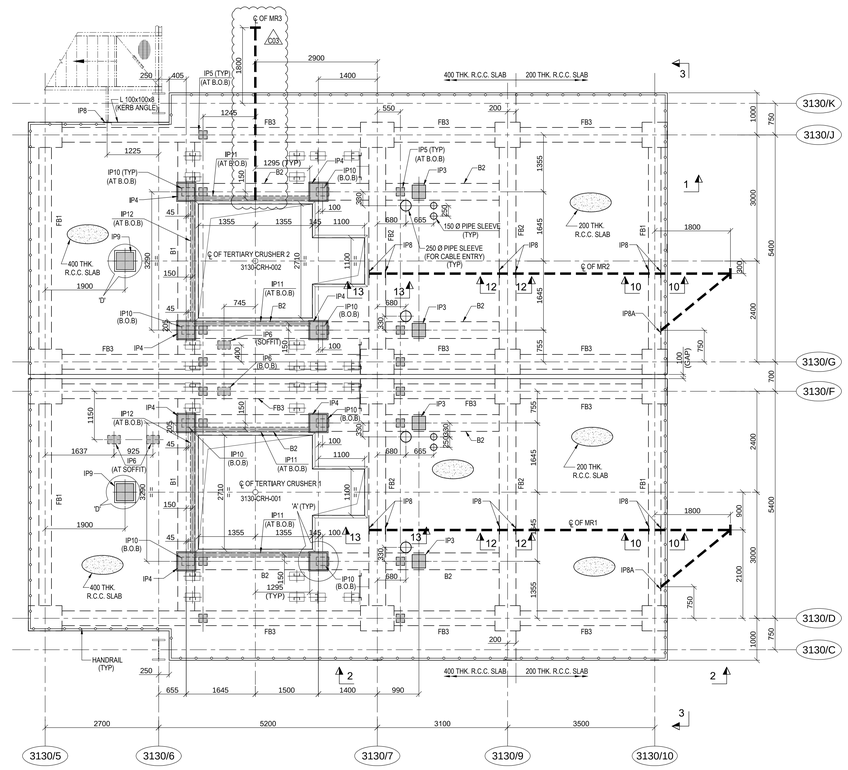
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DRAWING NUMBER: MD-216-3130-EG-CV-GAD-1007 | REV: C03



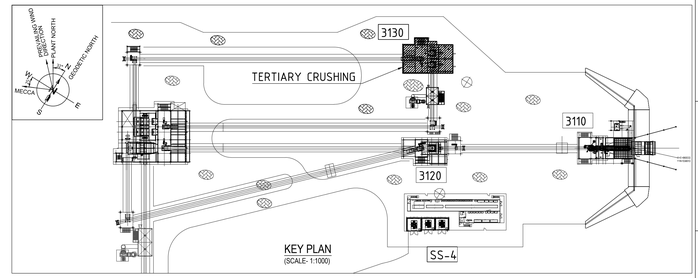
PLAN OF GRADE SLAB AT RL.913.000 (F.F.L) AND TIE BEAM AT RL. 912.850 (T.O.B)

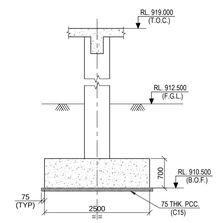
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PLAN AT RL. 919.000 (T.O.C)

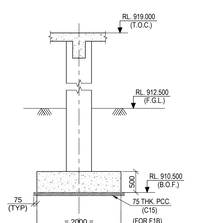
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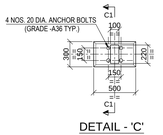
SECTION A-A

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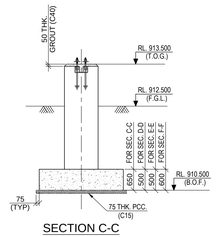
SECTION B-B

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DETAIL - 'C'

(SCALE-1:25)

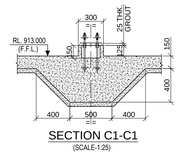


SECTION C-C
SECTION D-D
SECTION E-E
SECTION F-F

(SCALE - 1:50)

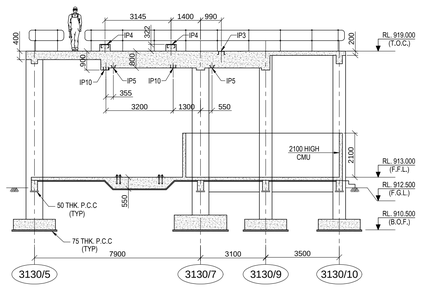
SCHEDULE OF BEAM

BEAM MKD.	SIZE		LEVEL
	WIDTH (mm.)	DEPTH (mm.)	
TB1	300	500	RL.+912.850 (T.O.B.)
TB2	300	550	
FB1	300	600	RL.+919.000 (T.O.C.)
FB2	400	900	
FB3	350	1100	
B1	400	900	
B2	400	800	



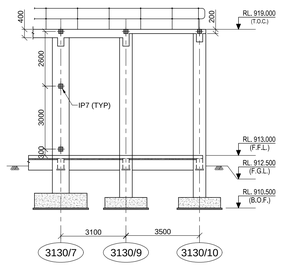
LAYOUT OF FOUNDATION AT RL. 910.500 (B.O.F)

(SCALE - 1:75)



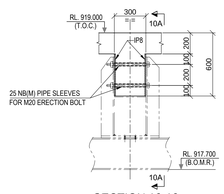
SECTION 1-1

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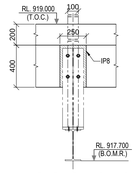
SECTION 2-2

(SCALE - 1:100)



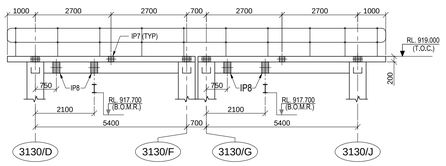
SECTION 10-10

(SCALE-1:20)



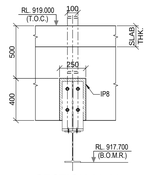
SECTION 10A-10A

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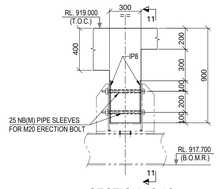
SECTION 3-3

(SCALE - 1:75)



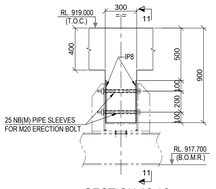
SECTION 11-11

(SCALE-1:20)



SECTION 12-12

(SCALE-1:20)



SECTION 13-13

(SCALE-1:20)

LEGENDS :-

RL. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.C. - TOP OF CONCRETE.
F.G.L. - FINISHED GROUND LEVEL.
B.O.F. - BOTTOM OF FOUNDATION
B.O.B. - BOTTOM OF BEAM
TYP. - TYPICAL
R.C.C - REINFORCED CEMENT CONCRETE
P.C.C - PLAIN CEMENT CONCRETE
MKD. - MARKED
THK. - THICK
MR. - MONORAIL
B.O.M.R. - BOTTOM OF MONORAIL
SEC. - SECTION
IP - INSERT PLATE
MM - MILLIMETER
SQ. - SQUARE
CMU - CONCRETE MASONRY UNIT

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE C35 FOR FOUNDATION.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDERED 500KN/SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS
   a) FOUNDATION = 75mm (BOTTOM & SIDE) 50 mm (TOP)
   b) TIE BEAM = 50 mm c) COLUMN = 50mm d) BEAM = 50mm e) SLAB = 40mm
6. FOR MINIMUM LAP LENGTH OF REINFORCEMENT REFER STANDARD DRAWING NO. MD-215-1000-EG-CV-DED-1001
7. THIS DRG. SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3130-EG-CV-GAD-1008 & MD-216-3130-EG-CV-DED-1007
8. FOR DETAIL OF HANDRAIL FIXING REFER STANDARD DRAWING.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-LPX-1009	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1010	TERTIARY CRUSHER HOUSE-PLAN VIEW	C03	17/02/2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C03 C02	HSK	SBY	RPK	AG	
MD-216-3130-EG-ME-LPX-1012	TERTIARY CRUSHER HOUSE-PLAN VIEW	C02	21/08/2020	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C02 C01	HSK	SBY	RPK	AG	
MD-216-3130-EG-ME-LPX-1013	TERTIARY CRUSHER HOUSE-PLAN VIEW	C01	09/04/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER COMMENTS DATED 04/04/2020 & REVISED AS MKD. C01	HSK	SBY	RPK	AG	
MD-216-3130-EG-ME-LPX-1014	TERTIARY CRUSHER HOUSE-PLAN VIEW				HSK	SBY	RPK	AG	
MD-216-3130-EG-ME-ELV-1017	TERTIARY CRUSHER HOUSE-ELEVATION	A02	12/03/2020	REVISED AS PER EMPLOYER COMMENTS ON 26/02/2020.	HSK	SBY	RPK	AG	
MD-216-3130-EG-ME-ELV-1018	TERTIARY CRUSHER HOUSE-ELEVATION	A01	27/01/2020	ISSUED OF REFERENCE	HSK	SBY	RPK	AG	

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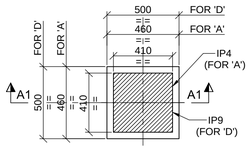
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LARSEN & TOUBRO DRAWING NUMBER	O1904.5_100_C_G_24065_001
REVISION	4
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31372
DOCUMENT ID	OU100670454
REVISION	4
DRAWN	HSK 10/01/2020
CHECKED	SBY 21/01/2020
ENGINEER	RPK 20/01/2020
APPROVED	AG 27/01/2020
EMPLOYER	
SCALE	1:100,1:75,1:50,1:25,1:20,1:10
	A0

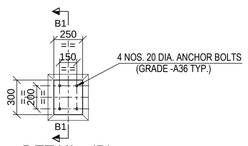
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
LAYOUT & DETAIL

DRAWING NUMBER: MD-216-3130-EG-CV-GAD-1007 | REV: C03

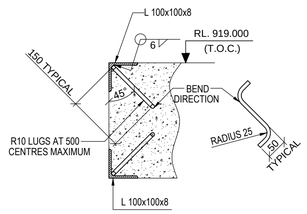
DRAWING NUMBER: MD-216-3130-EG-CV-GAD-1008 REV C03



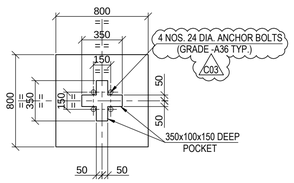
DETAIL 'A' & 'D'
(SCALE 1:20)



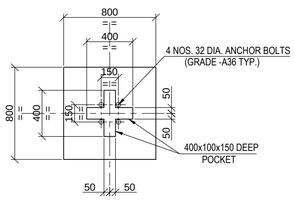
DETAIL - 'B'
(SCALE-1:25)



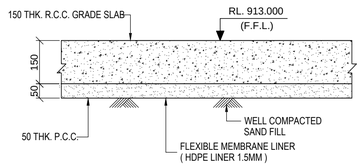
TYP. DET.OF KERB ANGLE
(FOR CUTOUT LOCATION)
(SCALE 1:10)



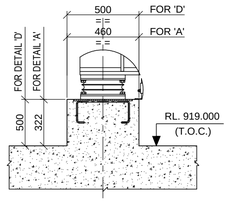
DETAIL OF COLUMN BASE
FOR PEDASTAL MKD. P1
(SCALE-1:25)



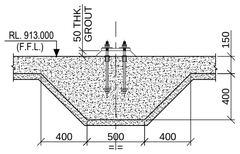
DETAIL OF COLUMN BASE
FOR PEDASTAL MKD. P2
(SCALE-1:25)



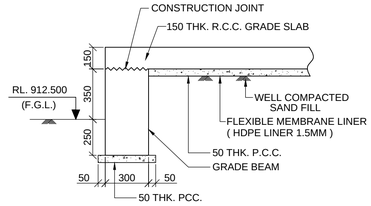
TYP. DET. OF 150 THK. GRADE SLAB
(SCALE 1:10)



SECTION A1-A1
(SCALE 1:20)

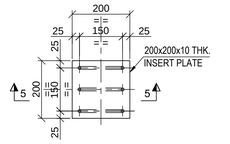


SECTION B1-B1
(SCALE-1:25)

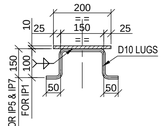


SECTION 4-4
(SCALE 1:20)

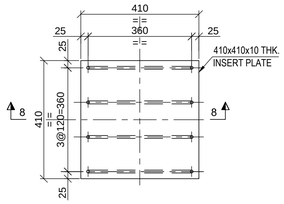
SCHEDULE OF INSERT PLATE		
INSERT PLATE MKD.	SIZE	PURPOSE
IP1	200x200x10 THK.	FOR EQUIPMENT
IP2	100x100x10 THK.	
IP3	320x320x10 THK.	
IP4	410x410x10 THK.	
IP9	410x410x10 THK.	
IP10	300x200x10 THK.	
IP11	2735x100x10 THK.	
IP12	2880x100x10 THK.	
IP5	200x200x10 THK.	FOR CABLE SUPPORT
IP6	300x200x10 THK.	FOR DUCT SUPPORT
IP7	200x200x10 THK.	FOR CABLE SUPPORT
IP8	400x250x10 THK.	FOR MONORAIL SUPPORT
IP8A	400x250x10 THK.	



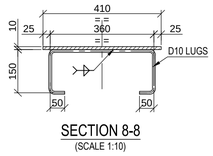
DETAIL OF INSERT PLATE - IP1,IP5,IP7
(SCALE 1:10)



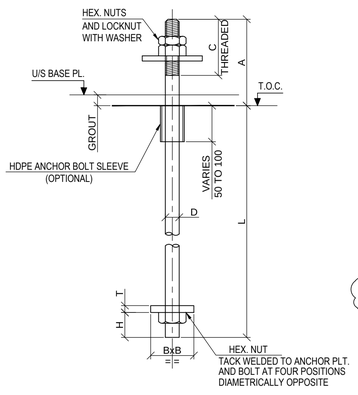
SECTION 5-5
(SCALE 1:10)



DETAIL OF INSERT PLATE - IP4 & IP9
(SCALE 1:10)



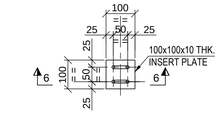
SECTION 8-8
(SCALE 1:10)



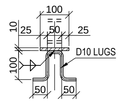
TYP. DETAIL OF BOLT

SCHEDULE OF BOLTS						
D BOLT DIA. MM	BxB BOTTOM PLATE (SQUARE) SQ MM	DIMENSIONS (MM)				
		L	A	C	T	H
20	60x60	450	160	110	10	30
24	75x75	600	175	125	12	30
32	100x100	800	190	140	16	35

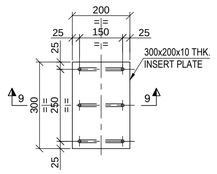
C03



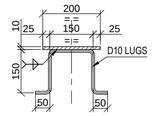
DETAIL OF INSERT PLATE - IP2
(SCALE 1:10)



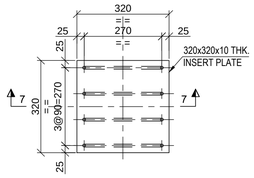
SECTION 6-6
(SCALE 1:10)



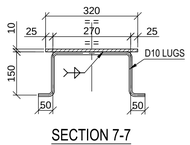
DETAIL OF INSERT PLATE - IP6 & IP10
(SCALE 1:10)



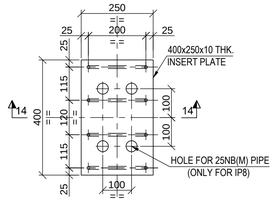
SECTION 9-9
(SCALE 1:10)



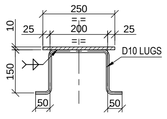
DETAIL OF INSERT PLATE - IP3
(SCALE 1:10)



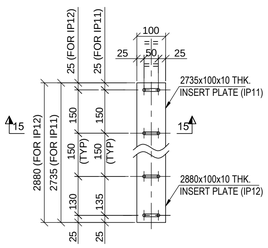
SECTION 7-7
(SCALE 1:10)



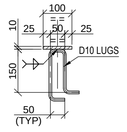
DETAIL OF INSERT PLATE - IP8 & IP8A
(SCALE 1:10)



SECTION 14-14
(SCALE 1:10)



DETAIL OF INSERT PLATE - IP11 & IP12
(SCALE 1:10)



SECTION 15-15
(SCALE 1:10)

LEGENDS :-

RL. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.C. - TOP OF CONCRETE.
F.G.L. - FINISHED GROUND LEVEL.
B.O.F. - BOTTOM OF FOUNDATION
B.O.B. - BOTTOM OF BEAM
TYP. - TYPICAL.
R.C.C - REINFORCED CEMENT CONCRETE
P.C.C - PLAIN CEMENT CONCRETE
MKD. - MARKED
THK. - THICK
MR. - MONORAIL
B.O.M.R. - BOTTOM OF MONORAIL
SEC. - SECTION
IP - INSERT PLATE
MM - MILLIMETER
SQ. - SQUARE

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF CONCRETE SHALL BE C35 FOR FOUNDATION.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615 - GRADE 60 (fy = 420 MPa).
4. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDERED 500KN/SQ. M.
5. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS a) FOUNDATION = 75mm (BOTTOM & SIDE) 50 mm (TOP) b) TIE BEAM = 50 mm c) COLUMN = 50mm d) BEAM = 50mm e) SLAB = 40mm
6. FOR MINIMUM LAP LENGTH OF REINFORCEMENT REFER STANDARD DRAWING NO. MD-216-1000-EG-CV-DED-1001
7. THIS DRG. SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3130-EG-CV-GAD-1007 & MD-216-3130-EG-CV-DED-1007
8. FOR DETAIL OF HANDRAIL FIXING REFER STANDARD DRAWING.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-LPX-1009	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1010	TERTIARY CRUSHER HOUSE-PLAN VIEW								
MD-216-3130-EG-ME-LPX-1012	TERTIARY CRUSHER HOUSE-PLAN VIEW	C03	07/01/2021	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C03	HSK	SBY	RPK	AG	
MD-216-3130-EG-ME-LPX-1013	TERTIARY CRUSHER HOUSE-PLAN VIEW	C02	21/08/2020	ISSUED FOR CONSTRUCTION & REVISED AS MKD. C02	HSK	SBY	RPK	AG	
MD-216-3130-EG-ME-LPX-1014	TERTIARY CRUSHER HOUSE-PLAN VIEW	C01	09/04/2020	ISSUED FOR CONSTRUCTION, REVISED AS PER EMPLOYER					
MD-216-3130-EG-ME-ELV-1017	TERTIARY CRUSHER HOUSE-ELEVATION			COMMENTS DATED 04/04/2020 & REVISED AS MKD. C01	HSK	SBY	RPK	AG	
MD-216-3130-EG-ME-ELV-1018	TERTIARY CRUSHER HOUSE-ELEVATION	A01	12/03/2020	ISSUED FOR REFERENCE	HSK	SBY	RPK	AG	

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Saudi Arabian Mining Company شركة معادن العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER: O19045_100_C_G_24065_002 — REVISION: 3
OUTOTEC DRAWING NUMBER: 900707 __GAU01 __ZC01 __MTL01 __31383 — DOCUMENT ID: OU100675149 — REVISION: 3

DRAWN	HSK	10/01/2020
CHECKED	SBY	21/01/2020
ENGINEER	RPK	20/01/2020
APPROVED	AG	12/03/2020
EMPLOYER		
SCALE	1:25,1:20,1:10	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
DETAILS & SECTIONS

DRAWING NUMBER: MD-216-3130-EG-CV-GAD-1008 — REV: C03

9. Appendix 2

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Subrata Dey

From: Joao Mendes <joao.mendes@sandvik.com>
Sent: 25 April 2022 18:43
To: Dhiman Majumdar; Vikash Sinha
Cc: Debiprasad Ghosh; Sonjit Mallick; Abhijit Ghosh; Ranjit Ghosh; Ashwin Prajapati; Mattias Lindh; B.Gangadhara Reddy; Subrata Dey; Jaydip Das; Gopinath T J
Subject: RE: Ma'aden MMGP Crushing System - Vibrations during No Load Trials of CH870i (Vertical Dynamic Load Application)

CAUTION: This is an external email. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Dear Sir,

Please see below in blue the feedback from R&D department.

- Confirmation for the vertical load application philosophy from Sandvik as given in the attachment or their suggested method
  - *The suggested methodology is reasonable.*
- The permissible RMS velocity below isolation pads i.e. top of insert plates.
  - *R&D doesn't have any recommendations on these values. Likely theses values should be stipulated in a relevant standard / design code for the foundation.*

Regards,

João Mendes

From: Dhiman Majumdar <dhimanm@lntecc.com>
Sent: Monday, April 25, 2022 11:33 AM
To: Vikash Sinha <vikash.sinha@sandvik.com>; Joao Mendes <joao.mendes@sandvik.com>
Cc: Debiprasad Ghosh <debiprasadghosh@lntecc.com>; Sonjit Mallick <sonjit@lntecc.com>; Abhijit Ghosh <abhijitg@lntecc.com>; Ranjit Ghosh <ranjitghosh@lntecc.com>; Ashwin Prajapati <ashwin.prajapati@sandvik.com>; Mattias Lindh <mattias.lindh@sandvik.com>; B.Gangadhara Reddy <bgreddy@lntecc.com>; Subrata Dey <subratadey@lntecc.com>; Jaydip Das <Jaydipdas@lntecc.com>; Gopinath T J <gopinathtj@lntecc.com>
Subject: Re: Ma'aden MMGP Crushing System - Vibrations during No Load Trials of CH870i (Vertical Dynamic Load Application)

Dear Sir/ Mr. Joao,

Awaiting response to L&T queries.

Please send it by today as our work cannot continue without Sandvik inputs.

Regards,
Dhiman

From: Dhiman Majumdar
Sent: Friday, April 22, 2022 4:47:16 PM
To: Vikash Sinha <vikash.sinha@sandvik.com>; Joao Mendes <joao.mendes@sandvik.com>
Cc: Debiprasad Ghosh <debiprasadghosh@lntecc.com>; Sonjit Mallick <sonjit@lntecc.com>; Abhijit Ghosh <abhijitg@lntecc.com>; Ranjit Ghosh <ranjitghosh@lntecc.com>; Ashwin Prajapati <ashwin.prajapati@sandvik.com>; Mattias Lindh <mattias.lindh@sandvik.com>; B.Gangadhara Reddy <bgreddy@lntecc.com>; Subrata Dey <subratadey@lntecc.com>; Jaydip Das <Jaydipdas@lntecc.com>; Gopinath T J <gopinathtj@lntecc.com>
Subject: RE: Ma'aden MMGP Crushing System - Vibrations during No Load Trials of CH870i (Vertical Dynamic Load Application)

Dear Joao,

Here is the load application philosophy for both vertical and horizontal direction. Bechtel has confirmed the horizontal load application philosophy and we need the following from Sandvik:

- Confirmation for the vertical load application philosophy from Sandvik as given in the attachment or their suggested method
- The permissible RMS velocity below isolation pads i.e. top of insert plates.

Regards,
Dhiman

-----Original Appointment-----
From: Dhiman Majumdar
Sent: 22 April 2022 12:17
To: Dhiman Majumdar; Vikash Sinha; Joao Mendes; Ashwin Prajapati; Mattias Lindh; B.Gangadhara Reddy; Subrata Dey; Jaydip Das; Gopinath T J
Cc: Debiprasad Ghosh; Sonjit Mallick; Abhijit Ghosh; Ranjit Ghosh
Subject: Ma'aden MMGP Crushing System - Vibrations during No Load Trials of CH870i (Vertical Dynamic Load Application)
When: 22 April 2022 14:00-14:30 (UTC+01:00) Brussels, Copenhagen, Madrid, Paris.
Where: Microsoft Teams Meeting

Dear Sirs,

Please find the invitation for the discussion.

Kindly share it with colleagues who are required to attend.

Microsoft Teams meeting

Join on your computer or mobile app
Click here to join the meeting

Join with a video conferencing device
748619195@t.plcm.vc
Video Conference ID: 134 021 397 7
Alternate VTC instructions

Learn More | Meeting options

Regards,
Dhiman

Sensitivity: LNT Construction Internal Use

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Appendix 3

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

BECHTEL

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

Outotec
LARSEN & TOUBRO

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-015619, (21-Oct-22

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 21 October 2022	Comment by	EPC Response DATE: 01 February 2024	Status	Remarks
			General Comments (cc)				
MD-216-3130-EG-CV-CAL-1008-A03	3130	1	Static analysis of the Tertiary Crusher support structure was carried out during the execution stage along with modal analysis to check the frequency separation from the exciting frequency of the cone crusher. Based on the investigation, the causes of excessive vibration felt on the support structure are: · Misinterpretation of the forced vibration nature of the equipment, · Incompliance with project requirements regarding being away from the range of excitation frequencies and vibration velocity amplitude limit; and · Forced vibration analysis of the structure to determine the vibration velocity was not performed earlier.	Mr. Hong Su	Noted.	Closed	